

Data Wrangling and Data Analysis

Data Visualization

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Weekly schedule for weeks 5, 6, 7, 8

Day	Time	Location	What
Monday/Tuesday/Wednesday	11:00 – 12:45	Andro C101/Ruppert blauw/Educatorium MEGATRON	Lecture (this)
Monday/Tuesday/Wednesday	15:15-17:00	BBG (check your group)	R Practical + questions
Friday	10:00 – 10:30	Online	Weekly exam

- **Please make sure you have installed:**
 - R
 - RStudio
 - ISLR package



Tue 4 Oct (*tomorrow*) and Wed 19 Oct

- **Your lecturer will be away**
- He will upload a video of the lecture which you can watch during or before the original lecture time
- **Two instructors will be available in the room from 12:00-13:00** to answer any questions you may have



Coordinators

Erik-Jan



Ayoub



Lecturer

Daniel



Tutors

Laura



Javier



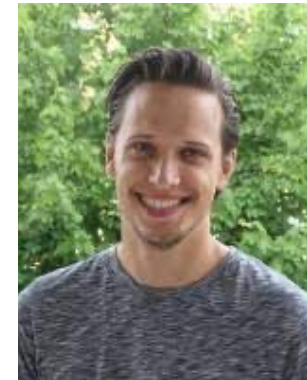
Mahdi



Anastasia



Jelle



Thom



Communication (see syllabus page 3)

- Please post questions about the **content** on Teams
- Questions on the lecture ;**CONTENT!** you do not want to post on Teams can be emailed to me
- Please **do not use Teams DM** for questions. These will get lost.
- Questions **about the course**: please email *course coordinator* (Hakim or Erik-Jan & Ayoub)
- Q on Practical & Assignment content: practical teachers (Laura, Javier, Mahdi, Anastasia, Jelle, Thom)

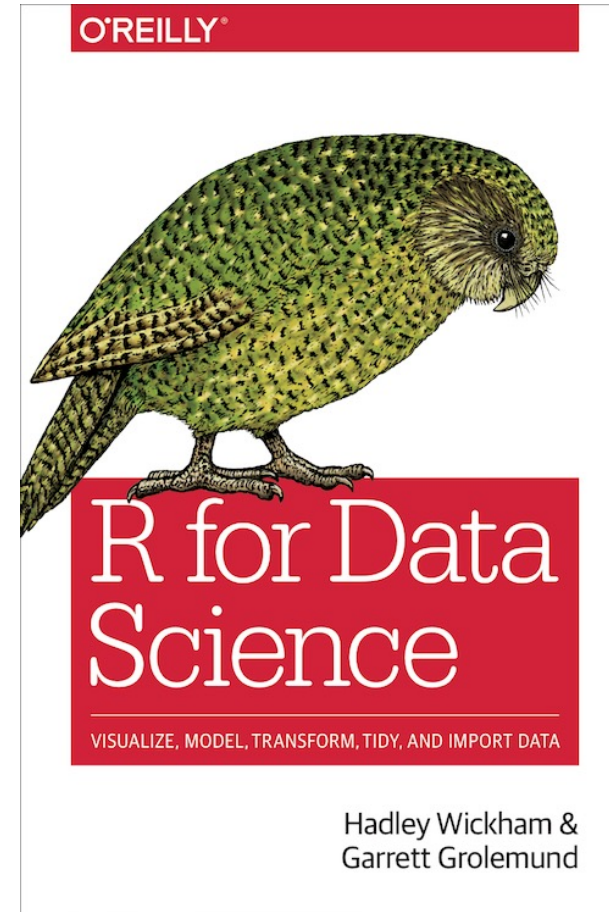


This week

- 1. Data visualization principles & Grammar of graphics**
2. Exploratory data analysis (EDA)
3. Building interactive dashboards using R Shiny

Reading materials for this week

- Chapters from **R for Data Science (R4DS)**, open access book at:
- <https://r4ds.had.co.nz>
- Today: ch 3 visualization
- Tomorrow: chh 3, 5, 7



Assignments this week

- Monday: Tidy data visualization with dplyr and ggplot
- Tuesday: Exploratory data analysis
- Wednesday: Dashboards in R Shiny



Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

Dressée par M. Minard, Inspecteur Général des Ponts et Chaussées en retraite Paris, le 20 Novembre 1869.

Les nombres d'hommes présents sont représentés par les largeurs des zones colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en travers des zones. Le rouge désigne les hommes qui entrent en Russie, le noir ceux qui en sortent. — Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M. M. Chiers, de Ségur, de Fezensac, de Chambray et le journal inédit de Jacob, pharmacien de l'Armée depuis le 28 Octobre.

Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Jérôme et du Maréchal Davoust qui avaient été détachés sur Minsk et Mohilow et ont rejoint vers Orscha et Witebsk, avaient toujours marché avec l'armée.

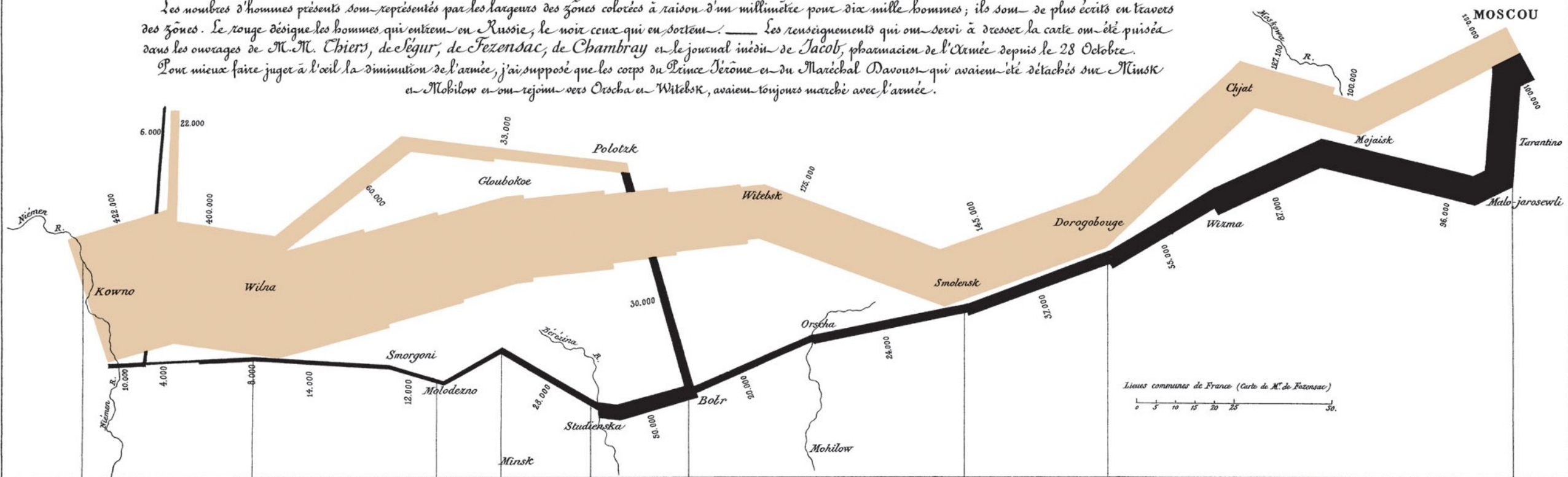
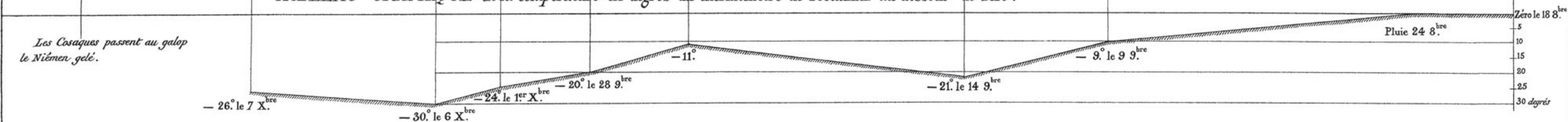


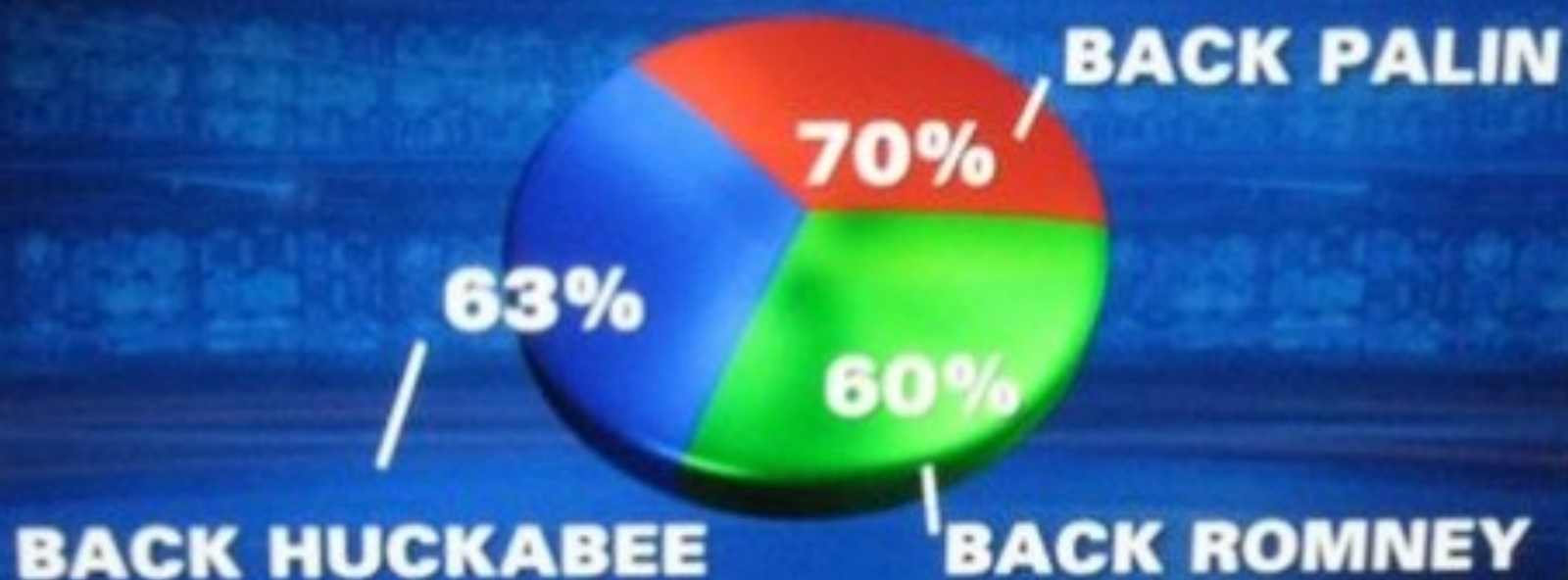
TABLEAU GRAPHIQUE de la température en degrés du thermomètre de Réaumur au dessous de zéro.



Les Cosaques passent au galop le Niemen gelé.

2012 PRESIDENTIAL RUN

GOP CANDIDATES



FOX

47'

SOURCE: OPINIONS

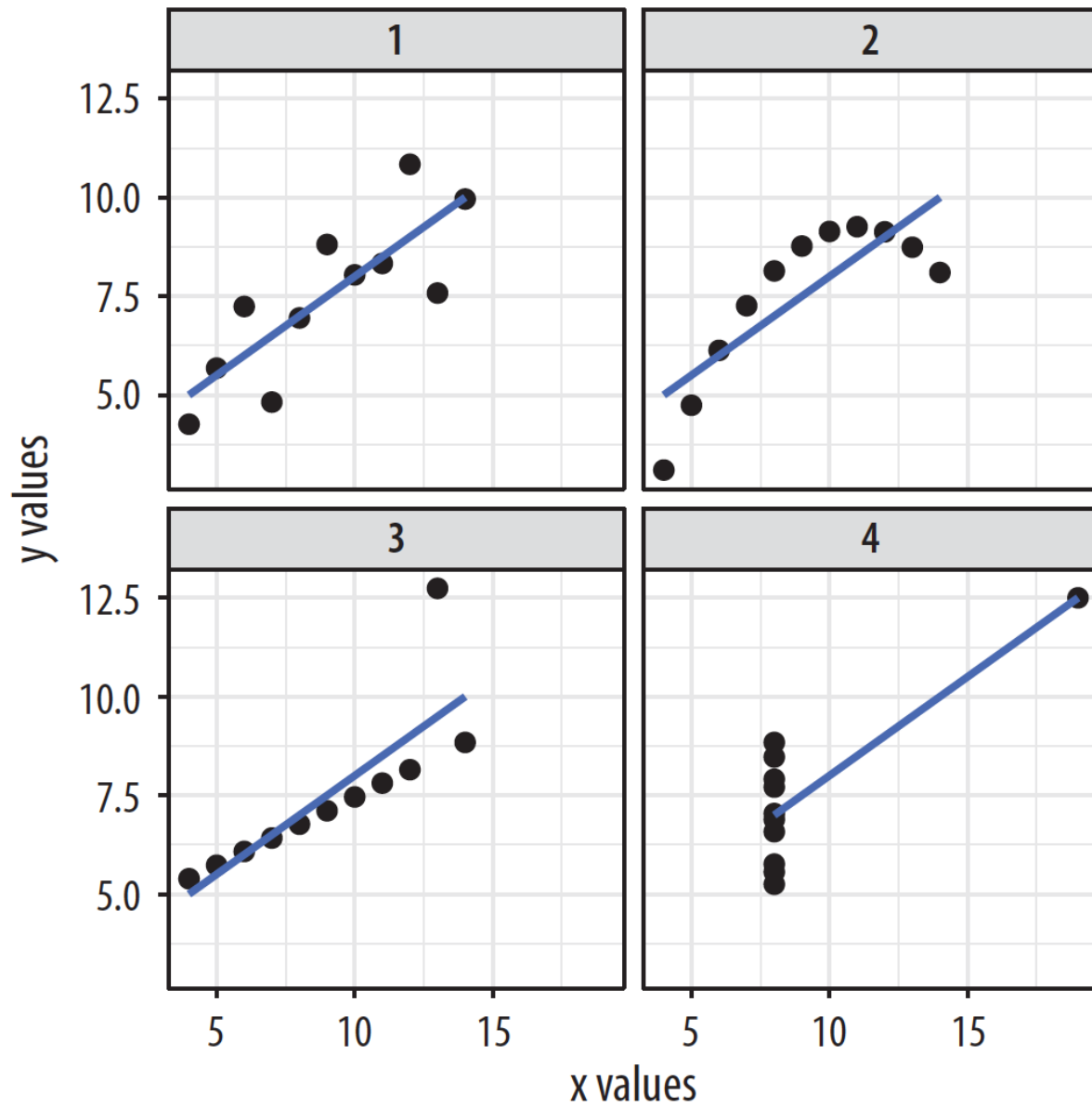
DYNAMIC

Today: visualization principles

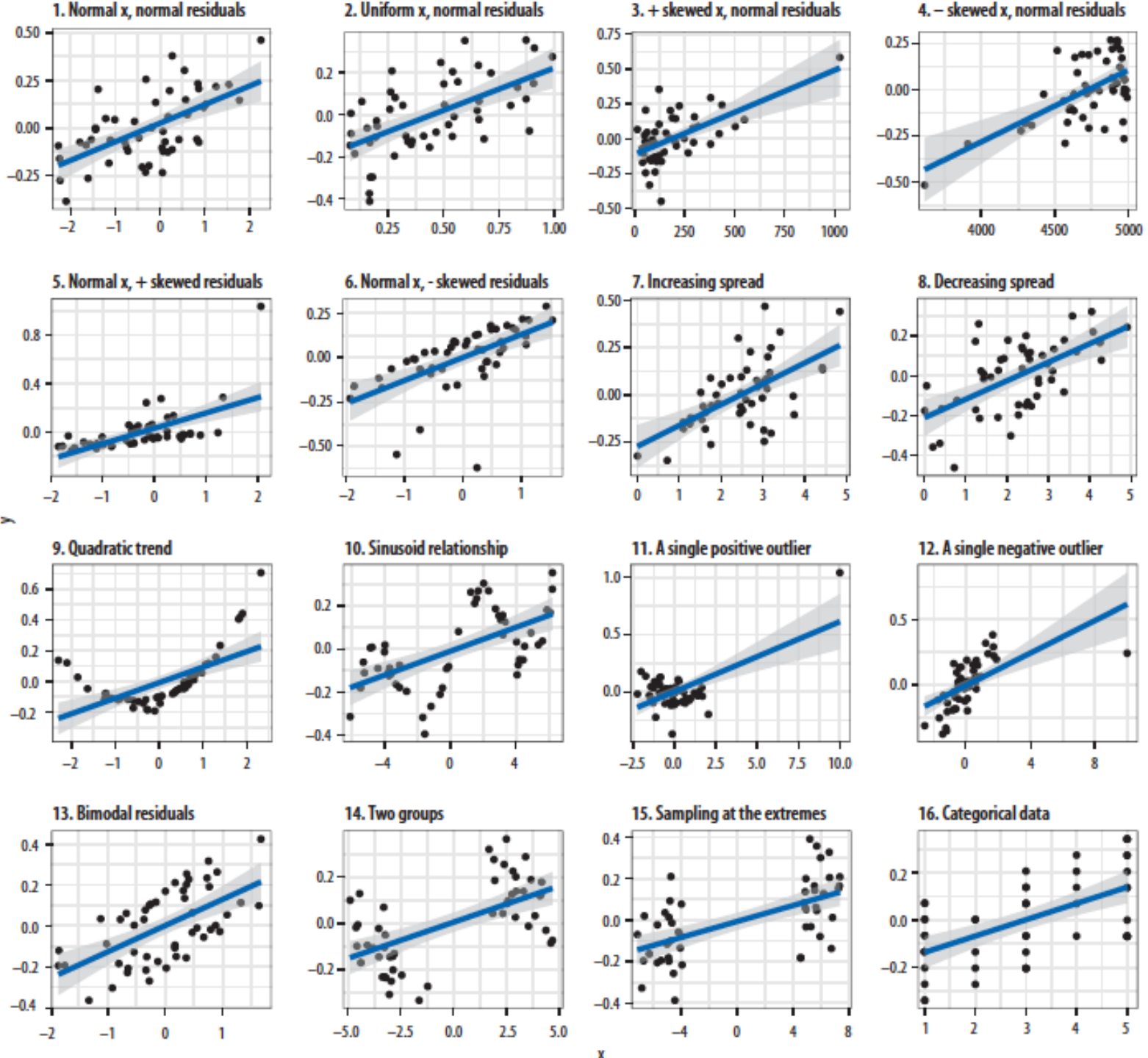
Data visualization

- **For exploration, data analysis** ←
- For communication
- For entertainment

Anscombe's quartet



Source: Healy (2019)



Graphics for data analysis

- The **human retina** can transfer around 10^6 or 10^7 bits per second to the brain;
- **Reading** transfers about 3 words, so $\sim 10^2$ or 10^3 bits/s;
- Potentially (!) visualization is about 4 orders of magnitude more powerful.

How can we leverage the human visual system to analyze data?



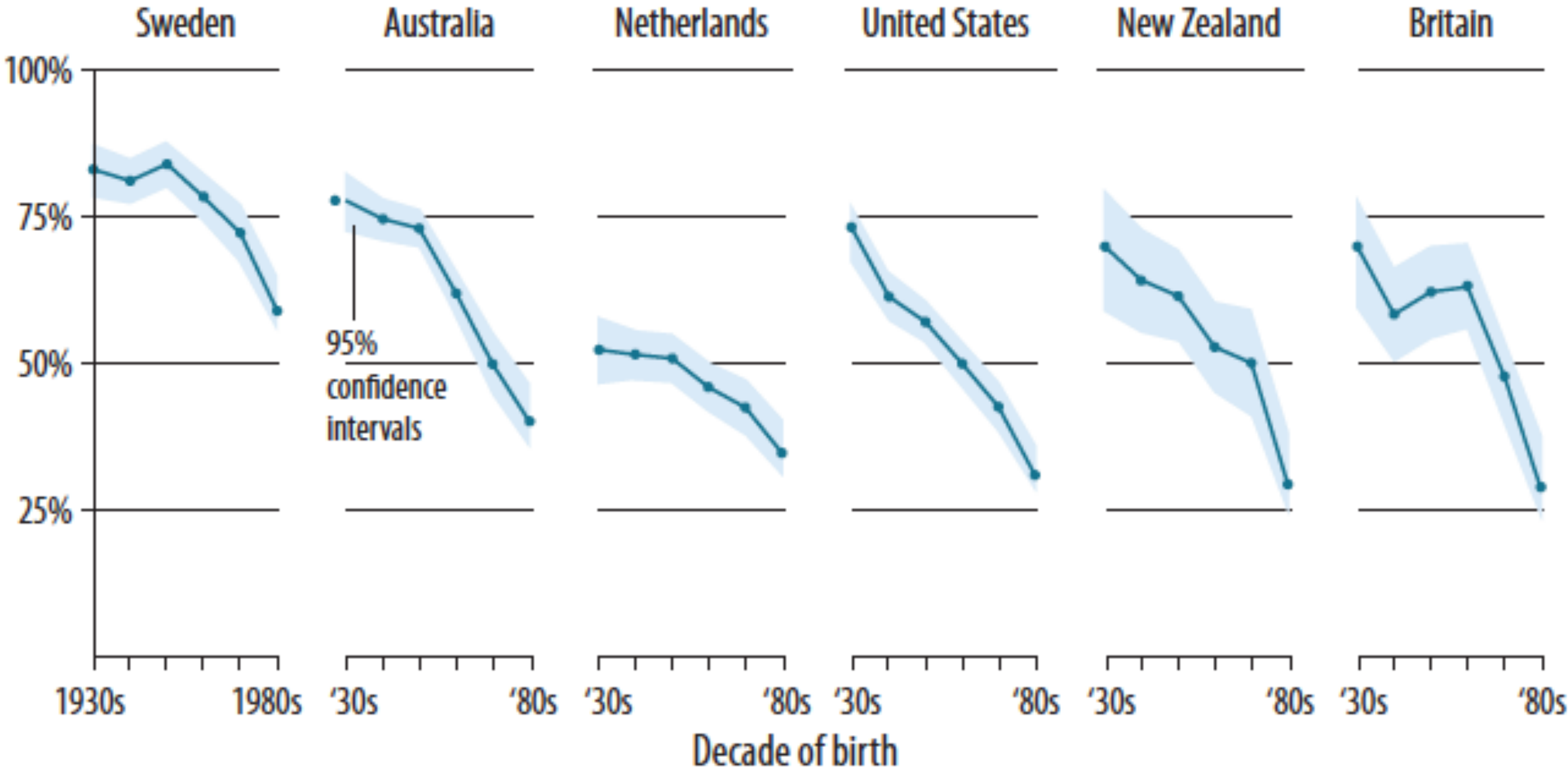
Plotting the right thing

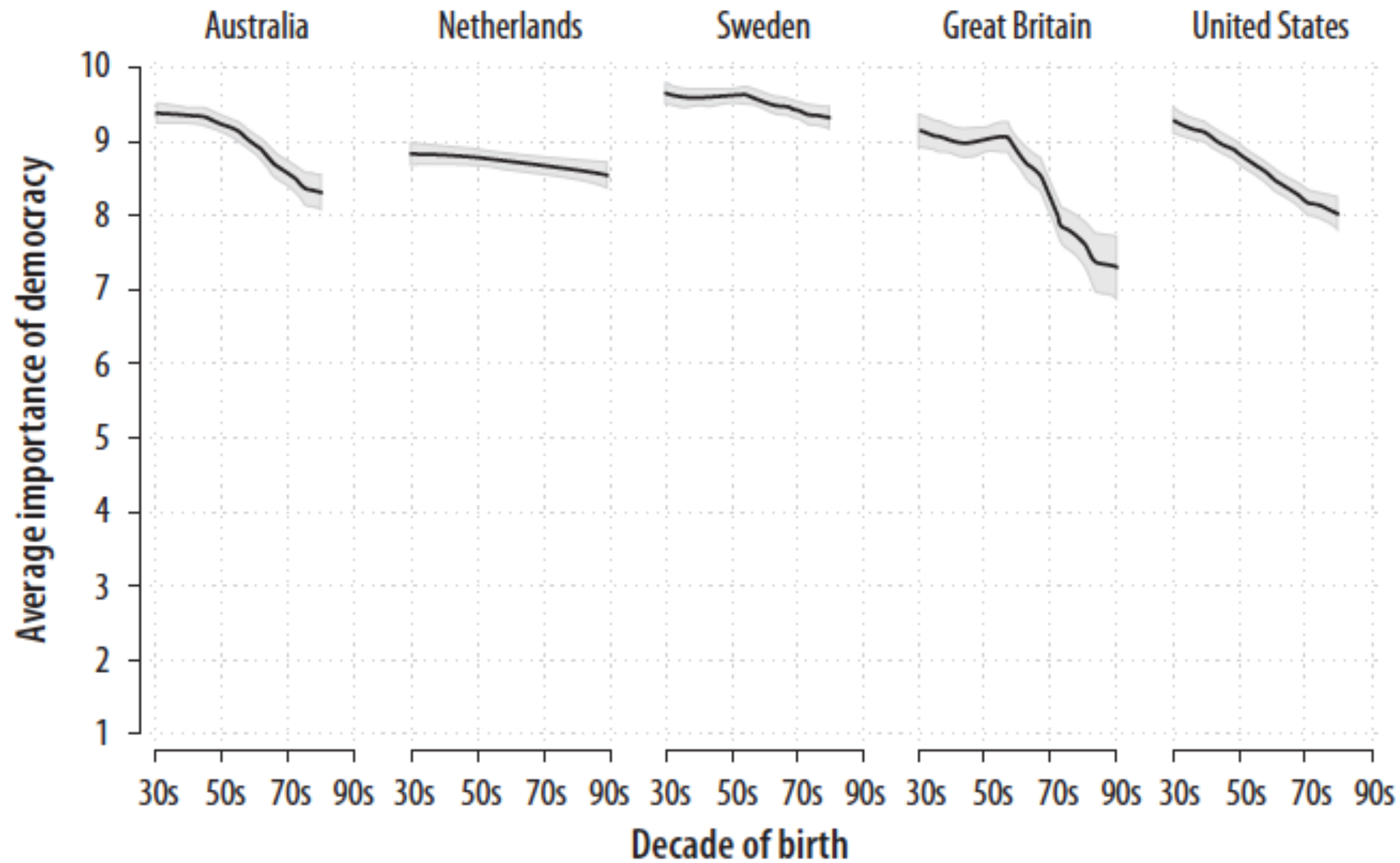
Most common problems:

- (Accidentally) misrepresenting what is being plotted
- Omitting baselines



Percentage of people who say it is "essential" to live in a democracy



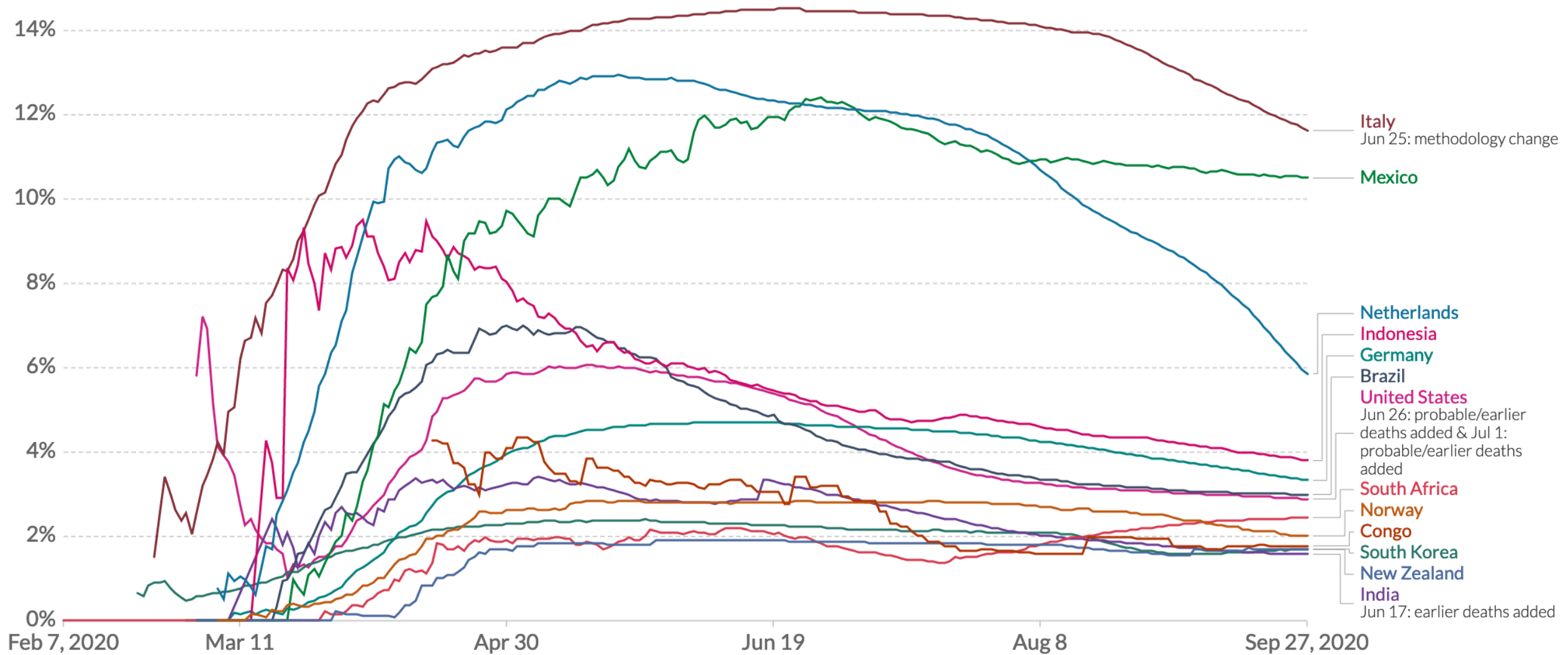


Graph by Erik Voeten, based on WVS 5

Source: Healy (2019)

Case fatality rate of the ongoing COVID-19 pandemic

The Case Fatality Rate (CFR) is the ratio between confirmed deaths and confirmed cases. During an outbreak of a pandemic the CFR is a poor measure of the mortality risk of the disease. We explain this in detail at OurWorldInData.org/Coronavirus

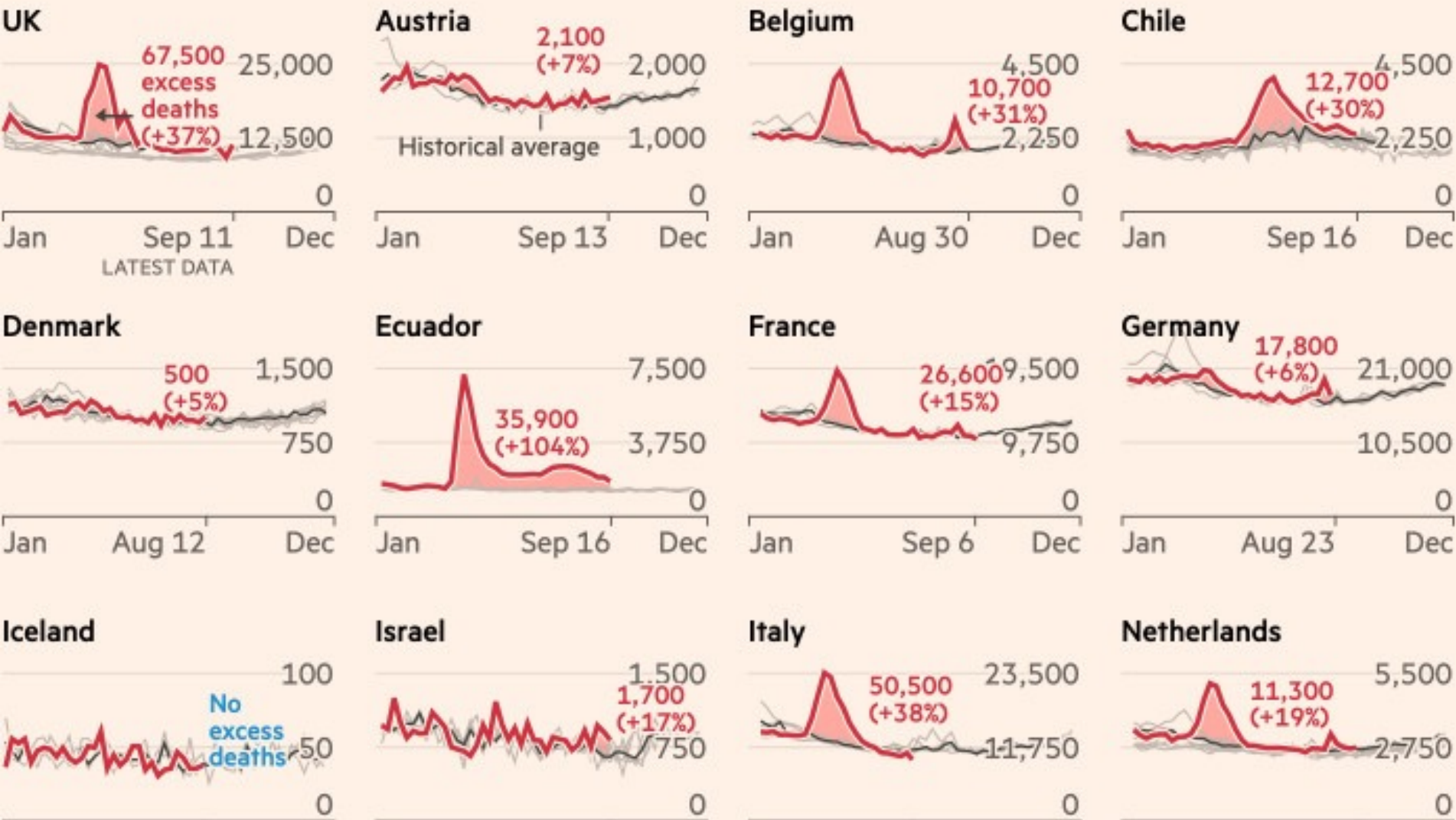


Death rates have climbed far above historical averages in many countries that have faced Covid-19 outbreaks

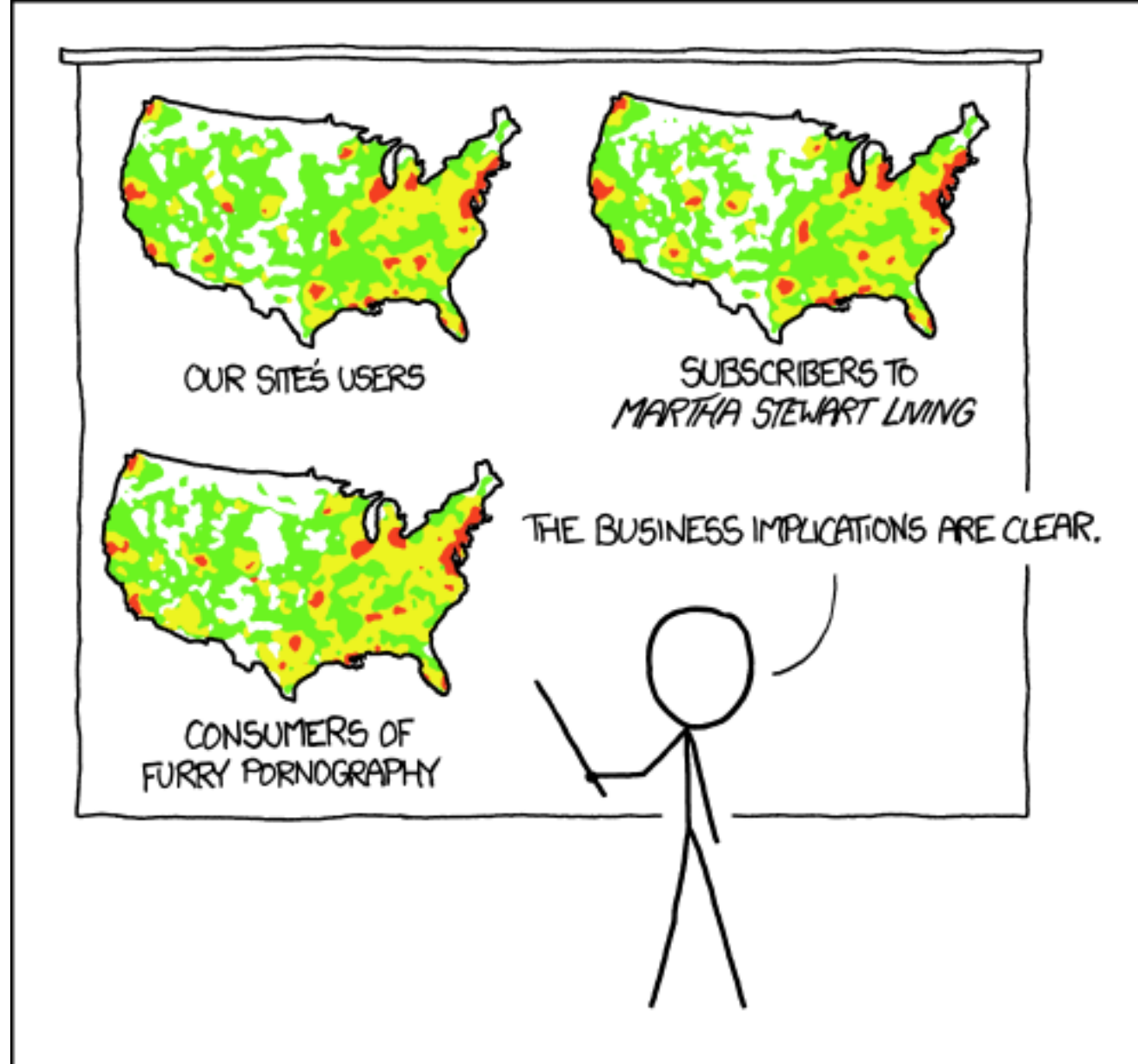
Good example (FT)

Number of deaths per week from all causes, 2020 vs recent years:

Shading indicates total excess deaths during outbreak



<https://www.ft.com/content/a298-5eb7-4633-b89c-cbdf5b38693>



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

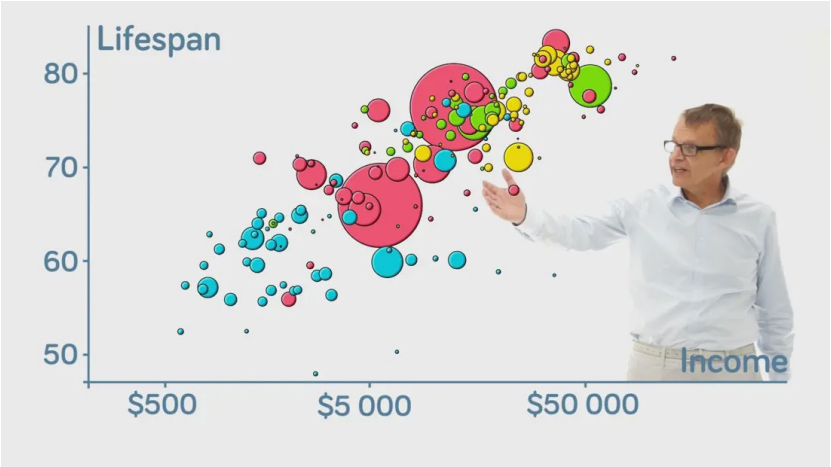


Data

Category	13-Mile race (min)	No. of runs	Mean lifespan (days + SEM)	P-value	Residual r ²
NI	0	7	152 ± 0.7	0.0134	4.0
	14	7	203 ± 0.3 ***	0.0001	7.0
	25	7	209 ± 0.3 **	0.0002	7.5
NI*	0	3	151 ± 0.8	<0.0001	17.3
	25	3	182 ± 0.2 ***	<0.0001	27.0
	25	3	182 ± 0.2 ***	<0.0001	27.0
NI* 2	0	4	171 ± 0.1	0.0005	2.3
	25	4	204 ± 0.2 ***	0.0001	1.0
NI* 3	0	4	124 ± 0.3 ***	<0.0001	4.0
	25	4	125 ± 0.3 ***	<0.0001	4.0
NI* 2,3	0	2	183 ± 0.2	0.0002	3.6
	25	2	204 ± 0.2 ***	0.0002	12.4
NI* 2,3	0	4	153 ± 0.3 ***	0.0001	8.6
	25	4	140 ± 0.3 ***	<0.0001	-8.5
NI* 2	0	3	181 ± 0.8	<0.0001	14.4
	25	3	203 ± 0.3 ***	<0.0001	12.1
NI* 1	0	3	203 ± 0.4 ***	<0.0001	
	25	3			



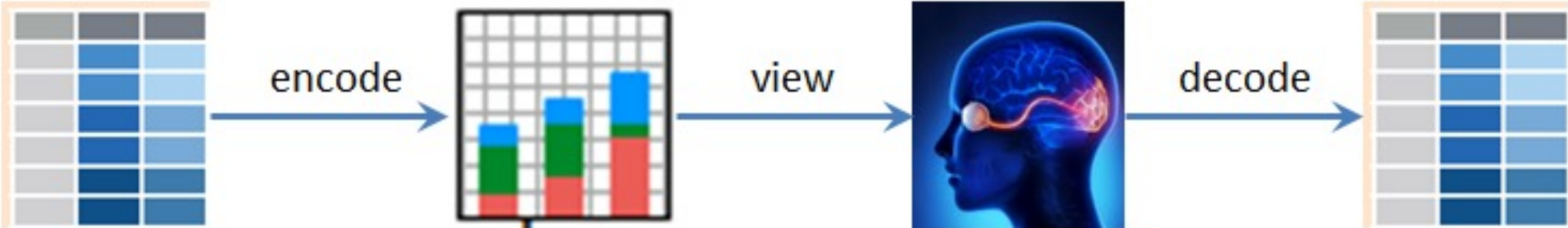
Picture



Brain



shutterstock.com · 1666544830



Making pictures that help analyze data

- We'd like to make, not just any kind of picture or graph, but one that transfers some part of the data to our brain
- How do we make sure that the graphs we make transfer:
 - The right part of the data, and;
 - As much of it as possible?

This is where the “**grammar of graphics**” comes in.

Goal is to **specify how data map to picture**, so the correct type and largest amount possible is transferred



Grammar of graphics (Wickham version)

- <http://r4ds.had.co.nz/visualize.html>
- Map raw data to following elements:
 - Aesthetics (position, shape, color, ...)
 - Geometric objects (points, lines, bars, ...)
 - Scales (continuous, discrete, ...)
 - Facets (small multiples)
- Additionally, can apply:
 - Statistical transformation (identity, binning, median, ...)
 - Coordinate system (Cartesian, polar, parallel, ...)



Grammar of graphics (Wickham version)

In R, grammar of graphics is implemented in `ggplot`, a function in the `ggplot2` package.



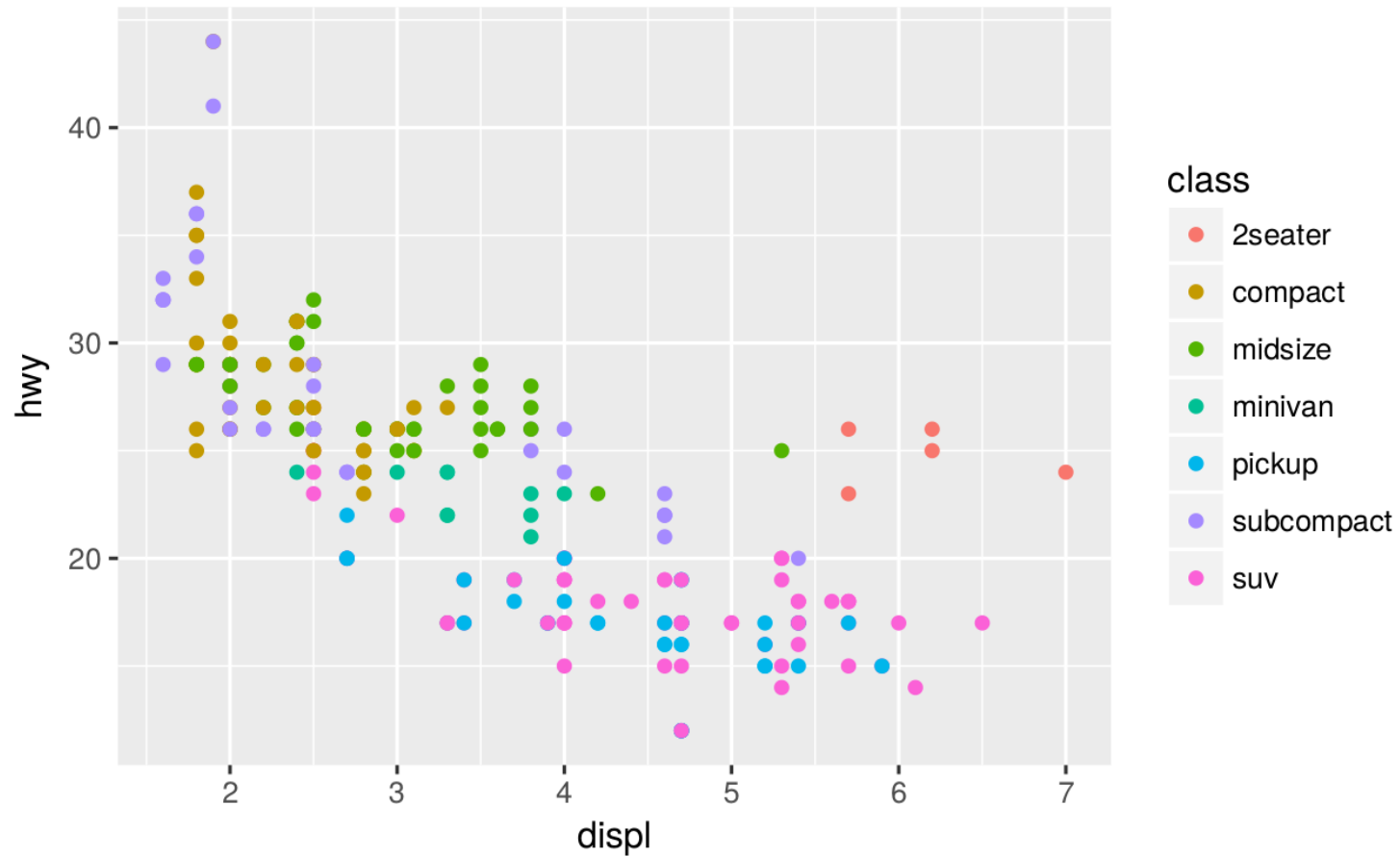
Example data set: cars

mpg

```
#> # A tibble: 234 × 11
#>   manufacturer model displ  year  cyl      trans  drv   cty   hwy fl
#>   <chr>    <chr> <dbl> <int> <int>    <chr> <chr> <int> <int> <chr>
#> 1      audi    a4    1.8  1999     4 auto(l5)   f    18    29   p
#> 2      audi    a4    1.8  1999     4 manual(m5)  f    21    29   p
#> 3      audi    a4    2.0  2008     4 manual(m6)  f    20    31   p
#> 4      audi    a4    2.0  2008     4 auto(av)   f    21    30   p
#> 5      audi    a4    2.8  1999     6 auto(l5)   f    16    26   p
#> 6      audi    a4    2.8  1999     6 manual(m5)  f    18    26   p
#> # ... with 228 more rows, and 1 more variables: class <chr>
```

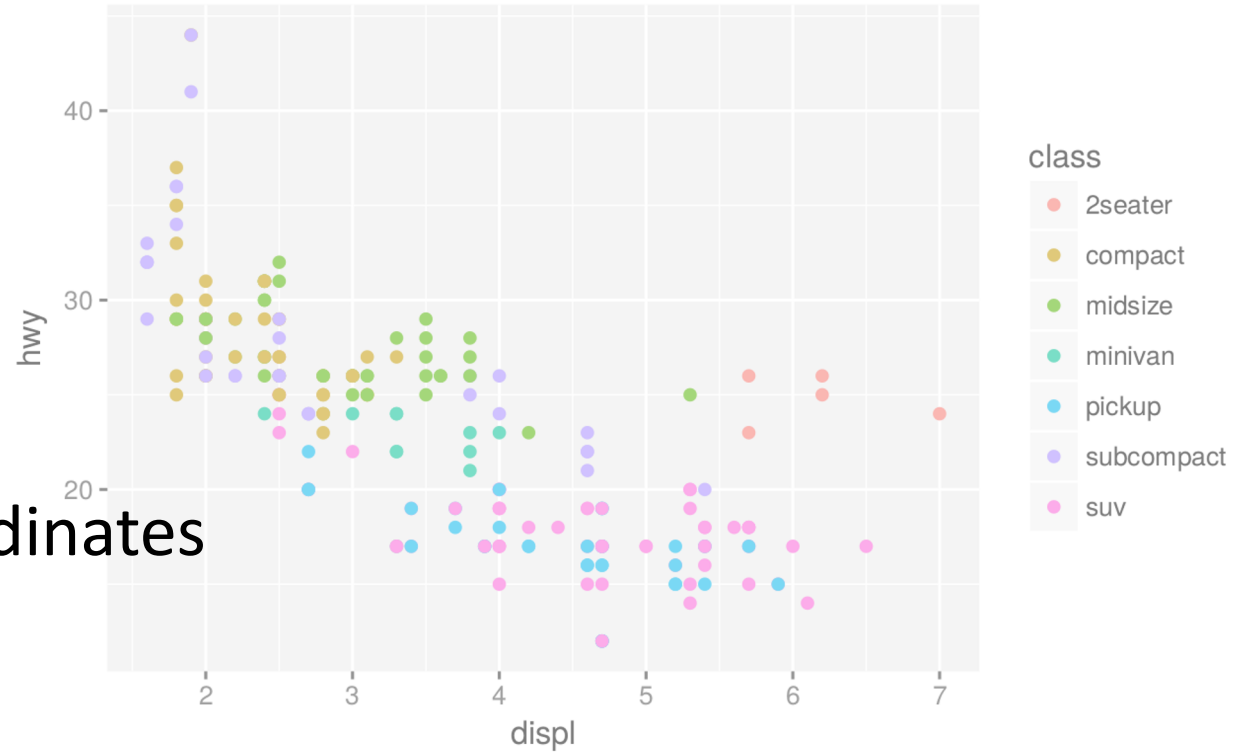


```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy,  
                           color = class))
```



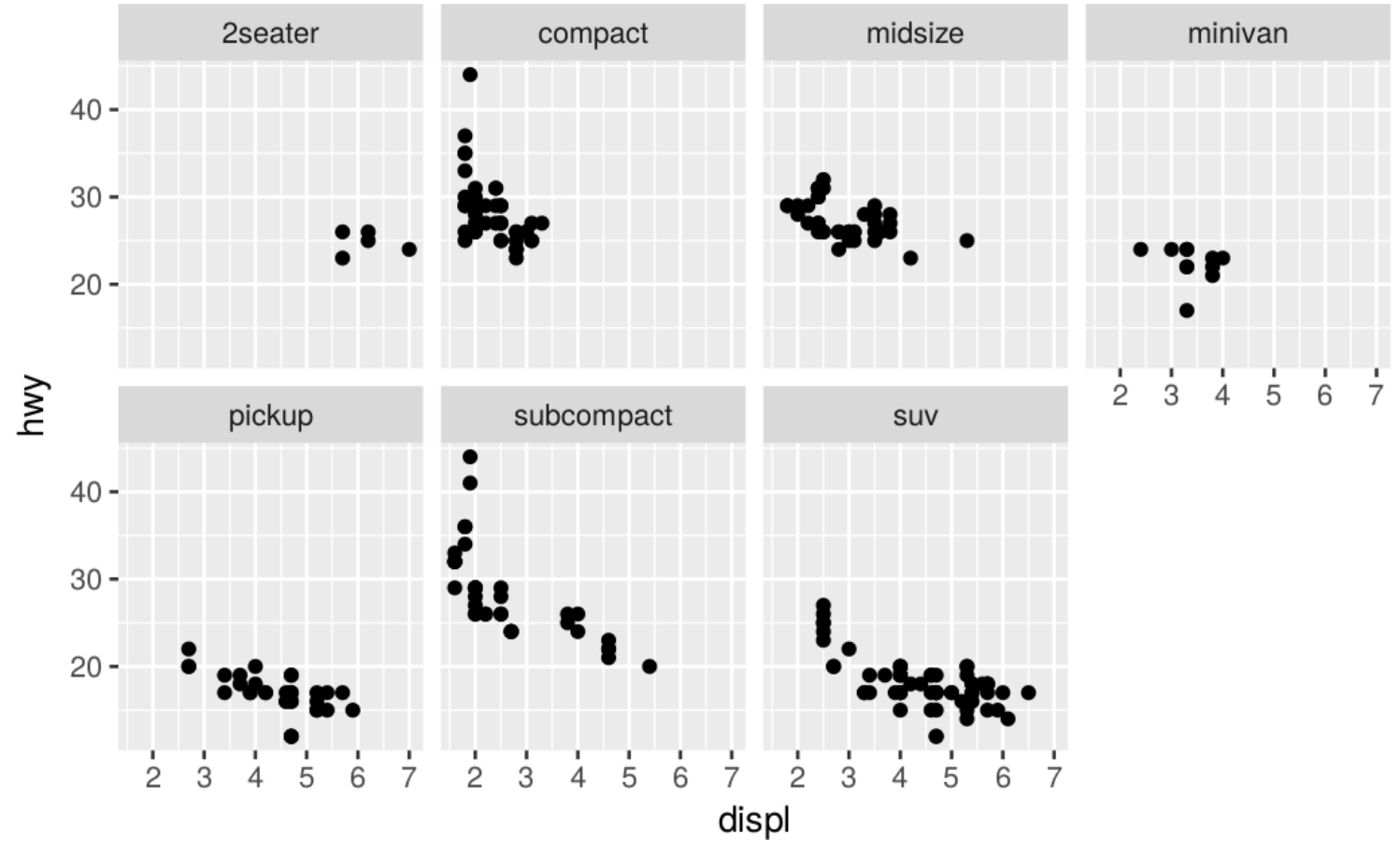
```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy,  
                           color = class))
```

- Aesthetics:
 - x-position mapped to engine size
 - y-position mapped to fuel efficiency
 - color mapped to car type
- Geometric objects: points
- Transformation: identity
- Scales: continuous, cartesian coordinates
- No facets



Facets

```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy)) +  
  facet_wrap(~ class, nrow = 2)
```



Transformation (stats)

1. **geom_bar()** begins with the **diamonds** data set

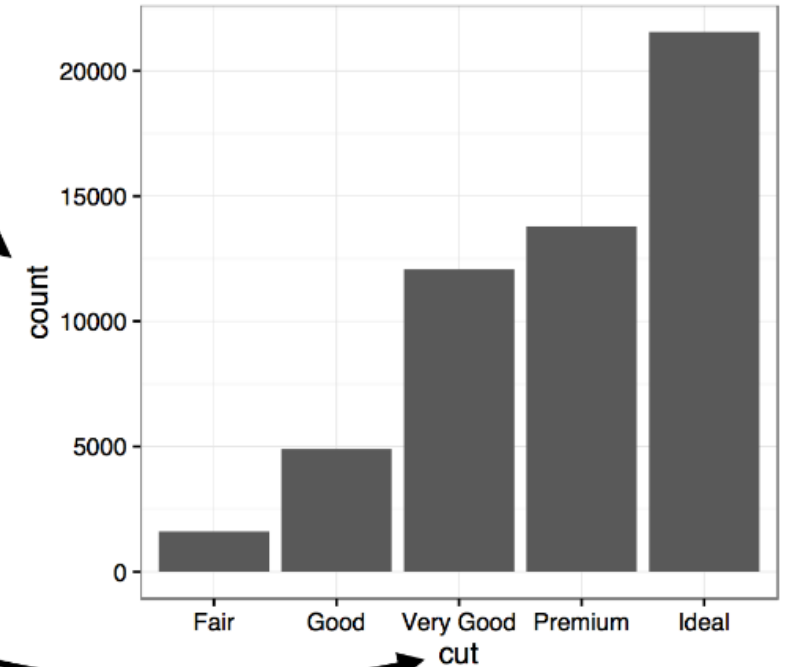
carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...

stat_count()

2. **geom_bar()** transforms the data with the "count" stat, which returns a data set of cut values and counts.

cut	count	prop
Fair	1610	1
Good	4906	1
Very Good	12082	1
Premium	13791	1
Ideal	21551	1

3. **geom_bar()** uses the transformed data to build the plot. cut is mapped to the x axis, count is mapped to the y axis.



What should I choose?

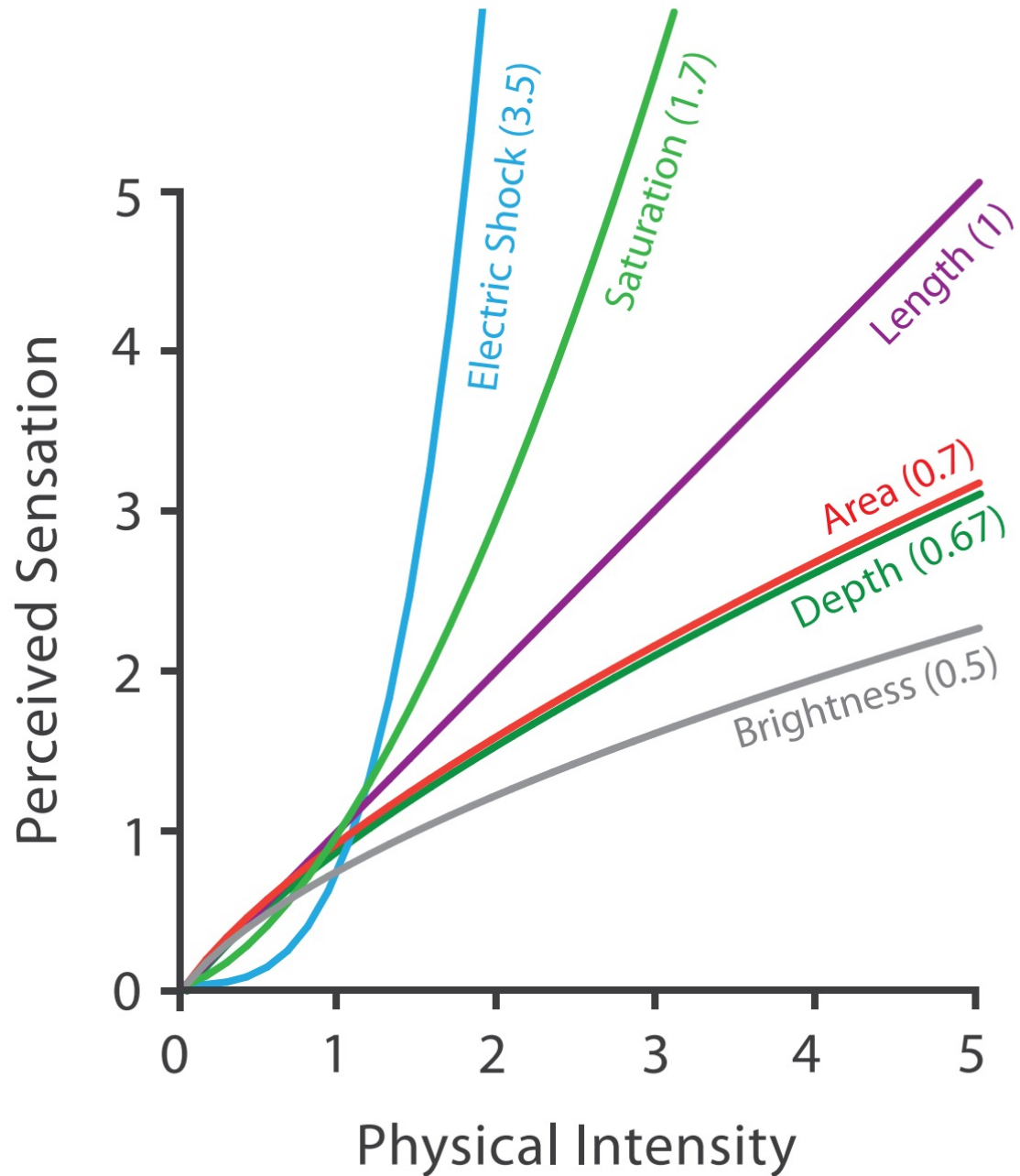
LES VARIABLES DE L'IMAGE

			POINTS			LIGNES			ZONES	
XY 2 DIMENSIONS DU PLAN										
Z										
TAILLE										
VALEUR										

LES VARIABLES DE SÉPARATION DES IMAGES

GRAIN								
COULEUR								
ORIENTATION								
FORME								

Steven's Psychophysical Power Law: $S = I^N$



Source: **Tamara Munzer** (2014). Visualization Analysis and Design.

Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude** Channels: **Ordered** Attributes

Position on common scale



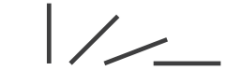
Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



Same

Same

Same

Most

Effectiveness

Least

➔ **Identity** Channels: **Categorical** Attributes

Spatial region



Color hue



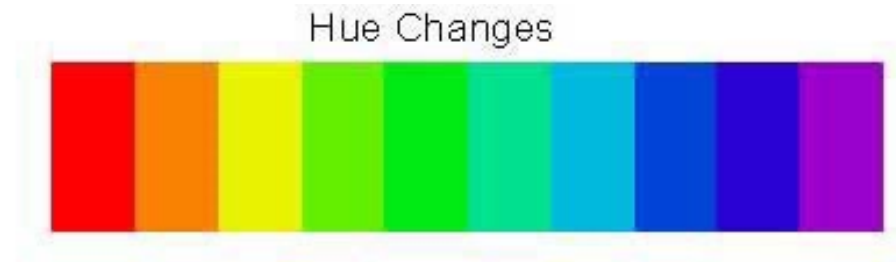
Motion

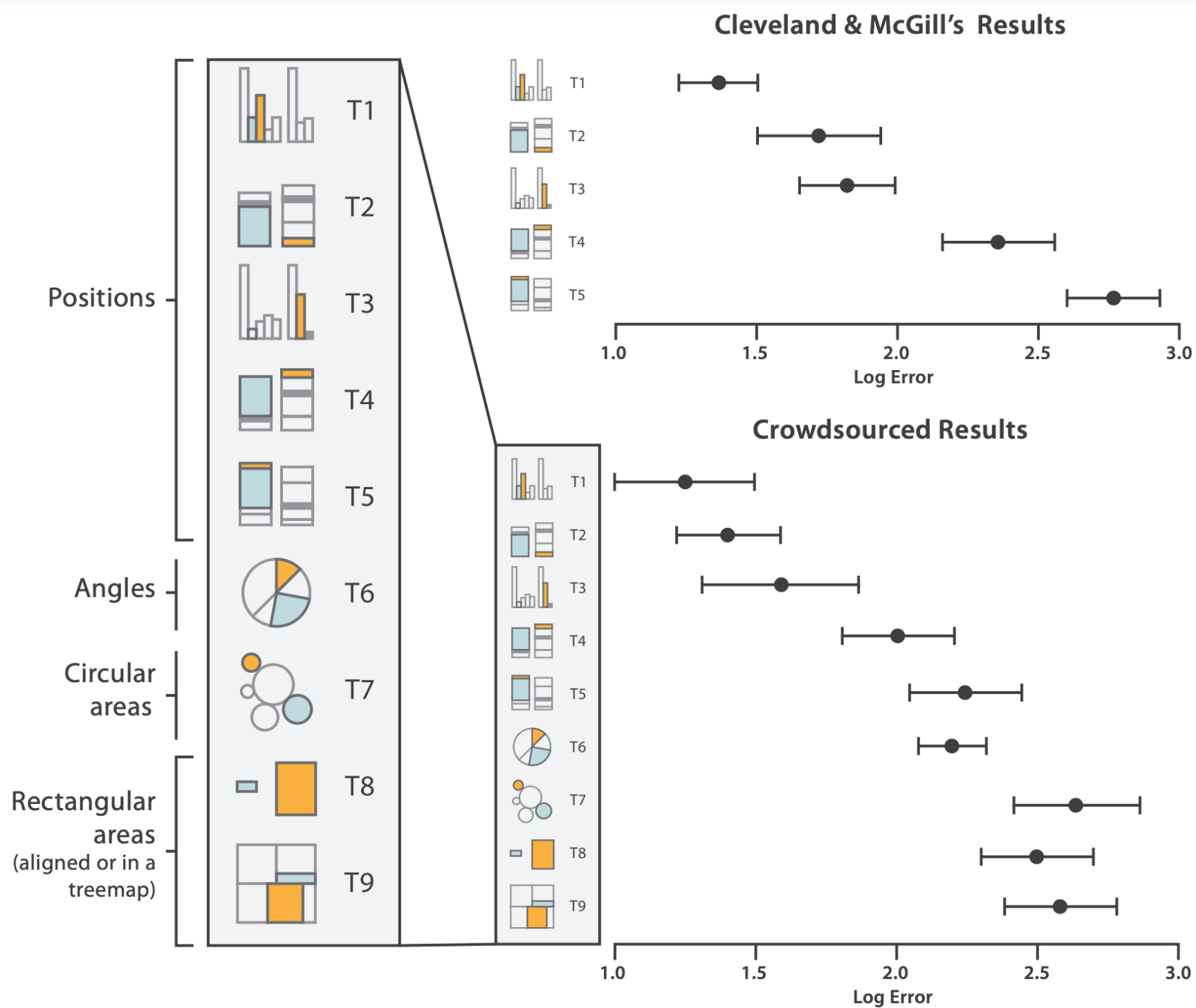


Shape



Color: hue-saturation-brightness (HSB)





Source: **Tamara Munzer** (2014).
Visualization Analysis and Design.

How many 5s in this display?

1561321203658413076510374627
4173127527327592732990709742
1703707774179527931749270973
4019743217909370945179279417

How many 5s in this display?

1561321203658413076510374627
4173127527|327592732990709742
1703707774179527931749270973
4019743217909370945179279417

Numerals differ only in **shape**, and are high-level symbols

You have to literally scan them **all** & count the 5s.

The distinction of **color** is immediate & **pre-attentive**

You only have to scan & count the 5s.

This is why **color** is an important visual attribute for a **categorical** variable in graphs

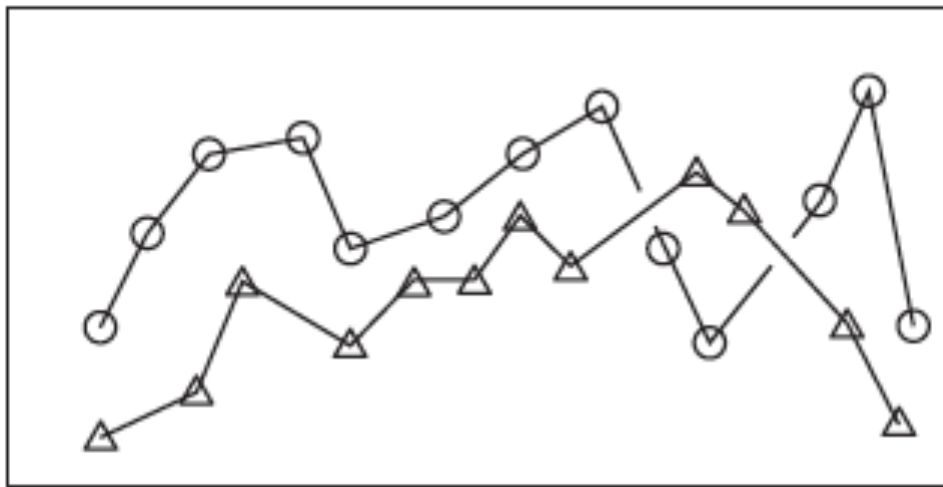
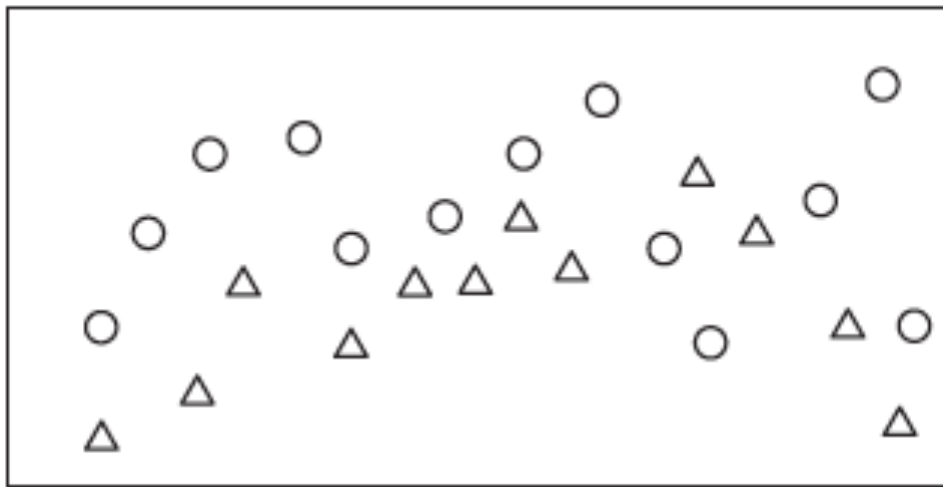
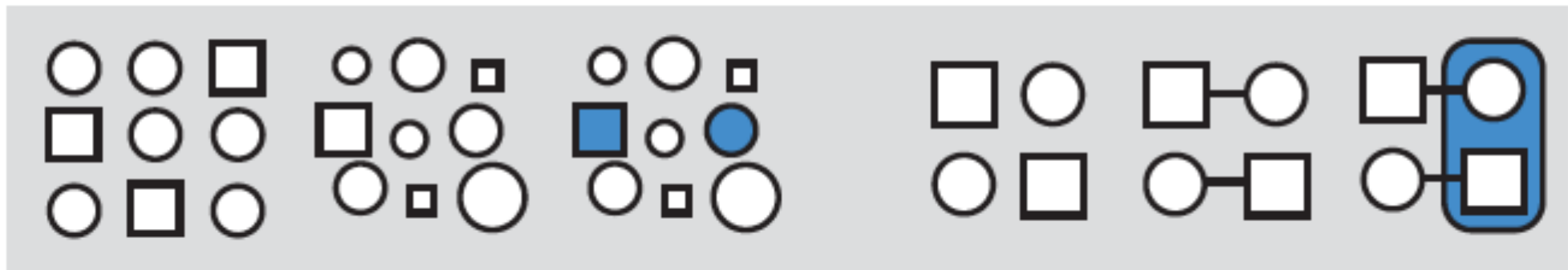
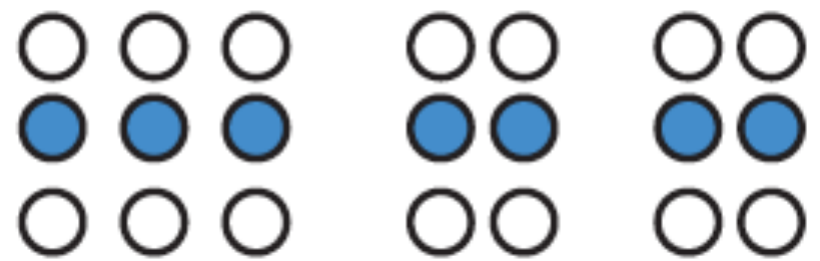
—



Gestalt principles of relatedness

- **Proximity:** Things that are spatially near to one another seem to be related.
- **Similarity:** Things that look alike seem to be related.
- **Connection:** Things that are visually tied to one another seem to be related.
- **Continuity:** Partially hidden objects are completed into familiar shapes.
- **Closure:** Incomplete shapes are perceived as complete.
- **Figure and ground:** Visual elements are taken to be either in the foreground or in the background.
- **Common fate:** Elements sharing a direction of movement are perceived as a unit.

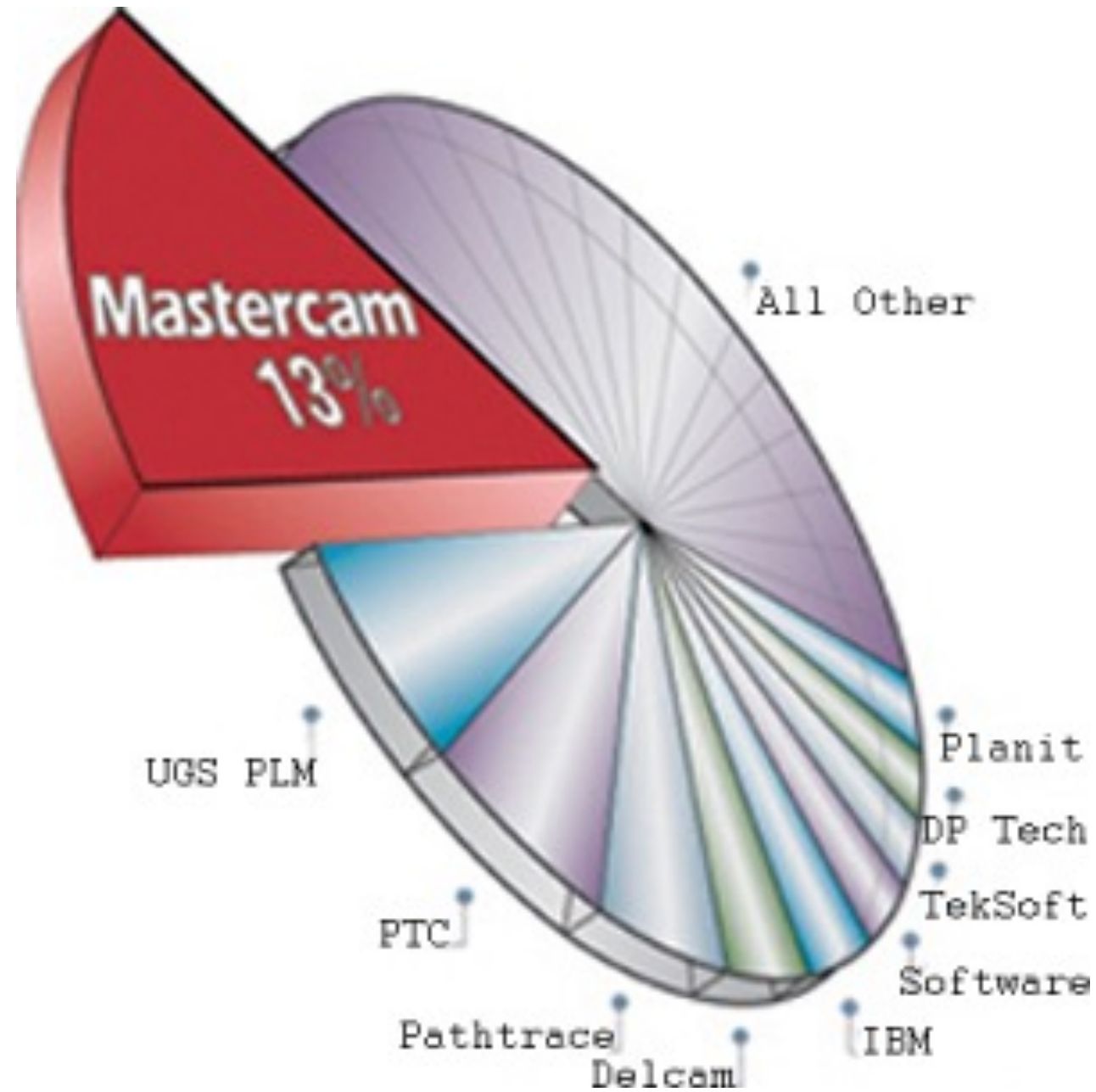




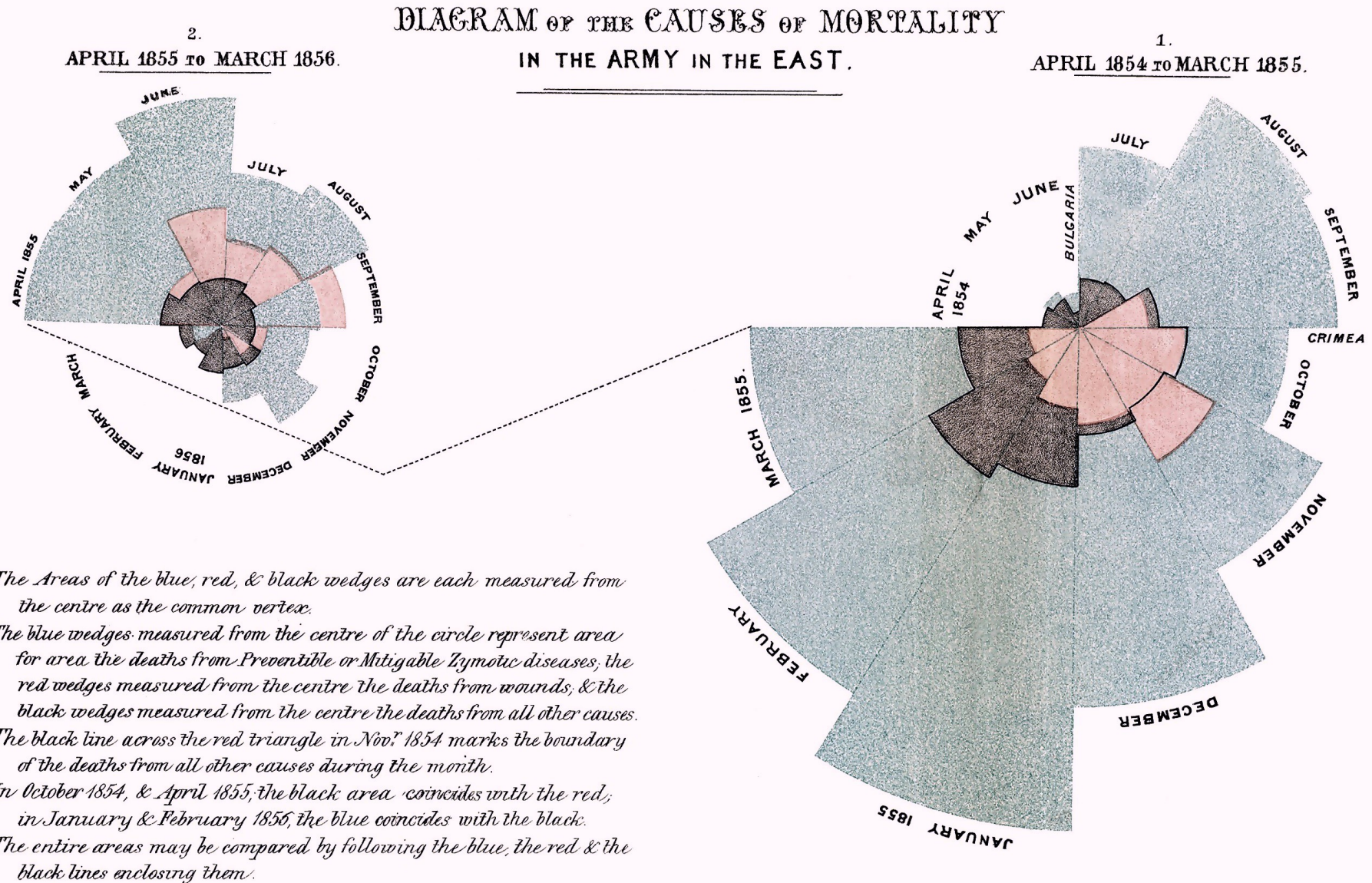
Some (distilled) principles from Tufte

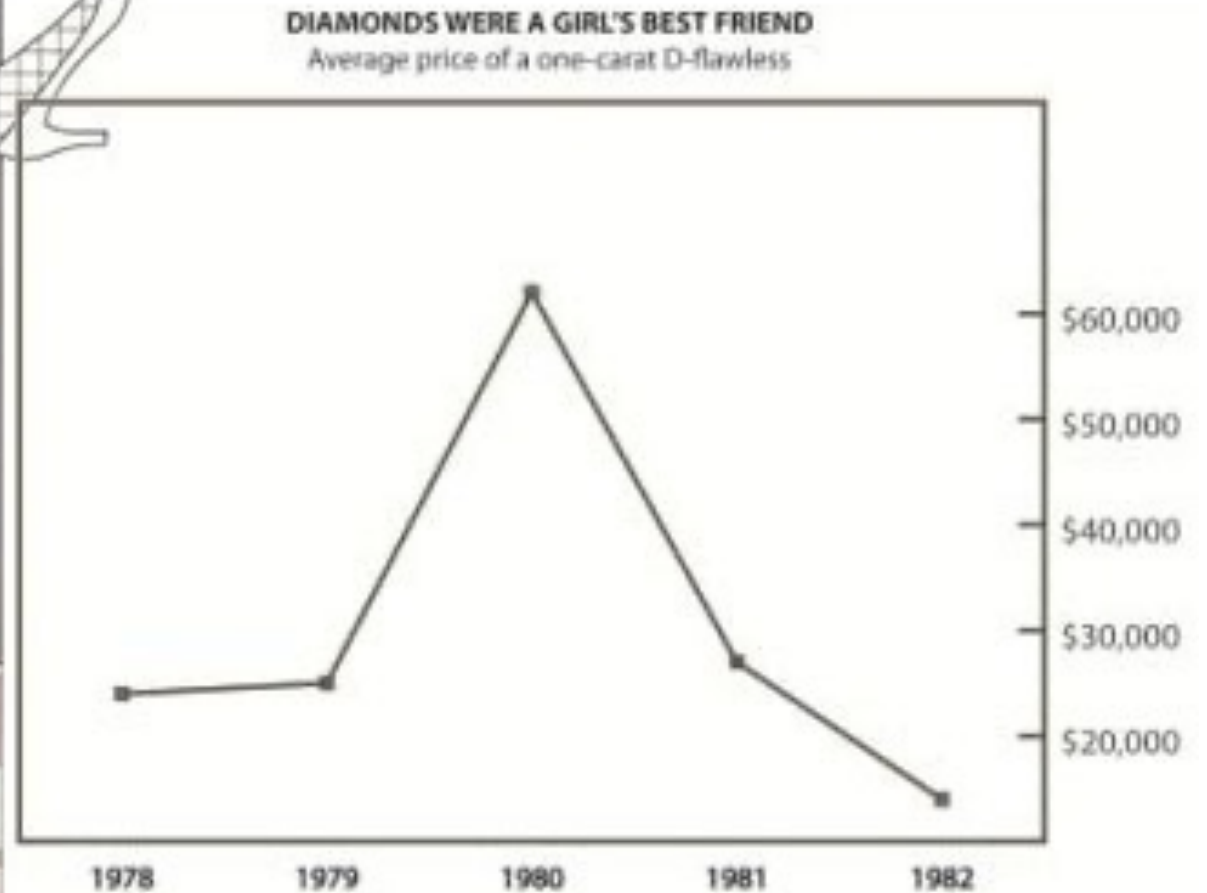
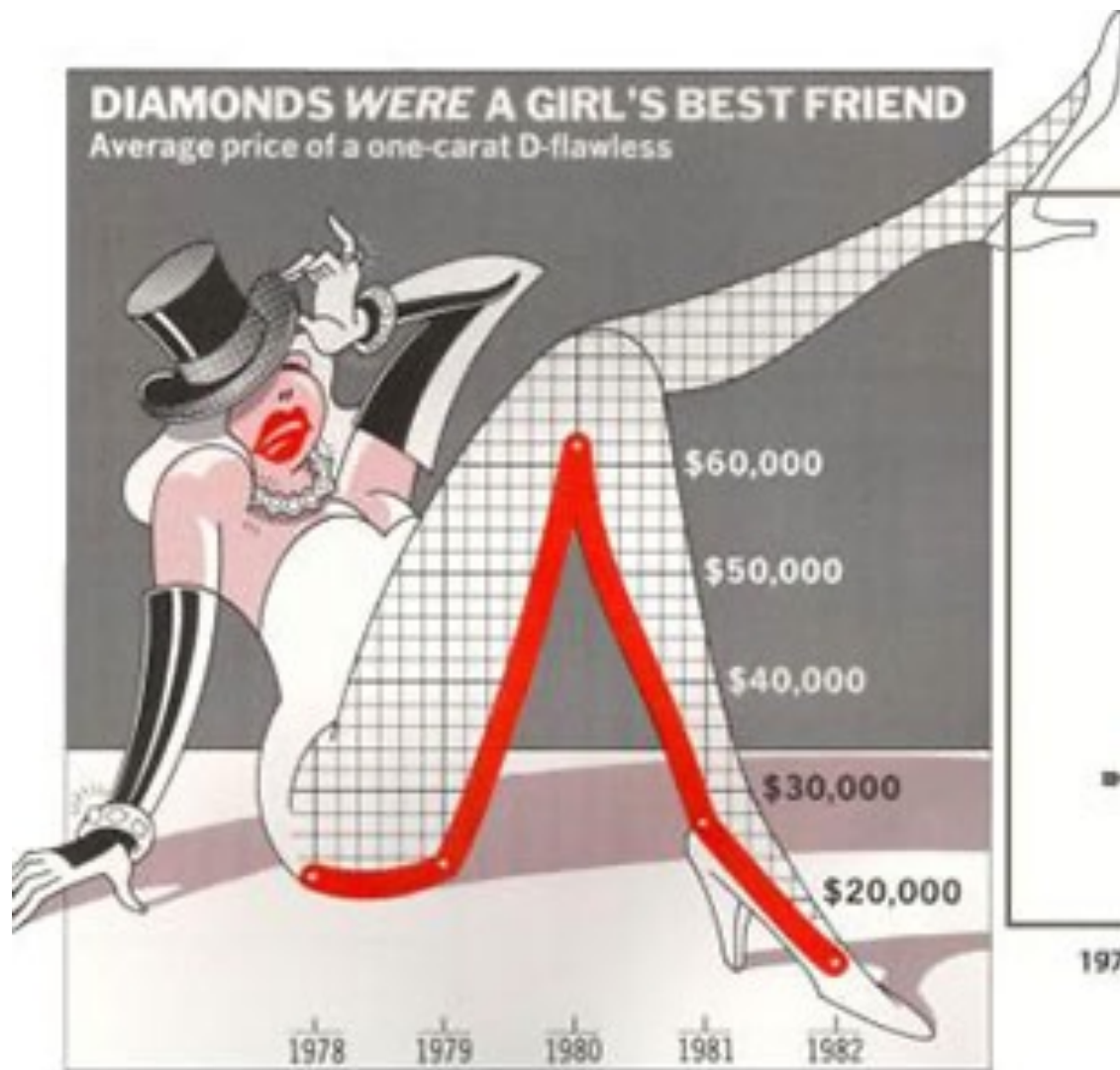


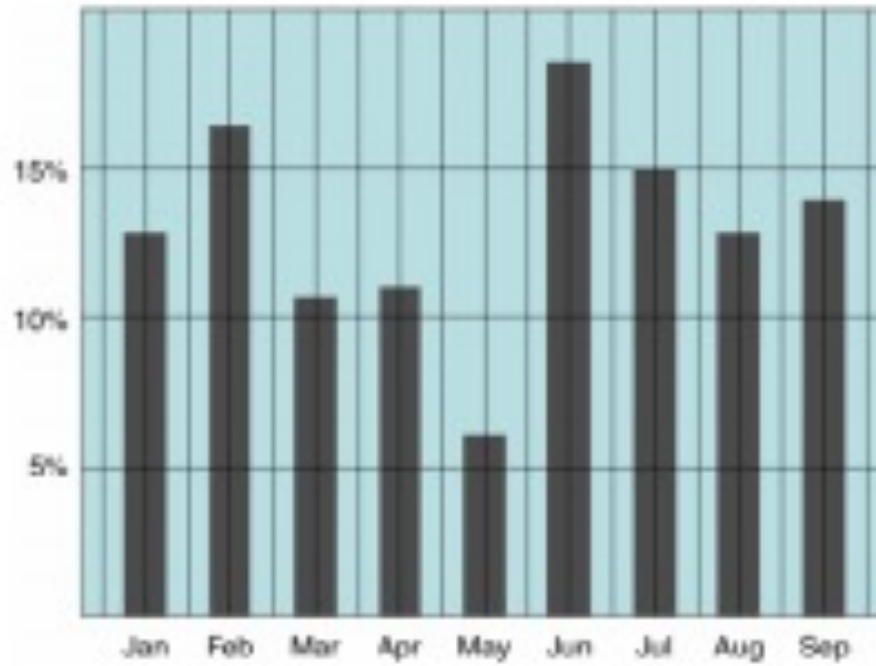
- Ask how data maps to perception
- Ask which comparisons you want, guide eye to those
- Maximize data-to-ink ratio
- Present more data (without losing interpretability)
- Use levels of detail
- (Remember narrative)



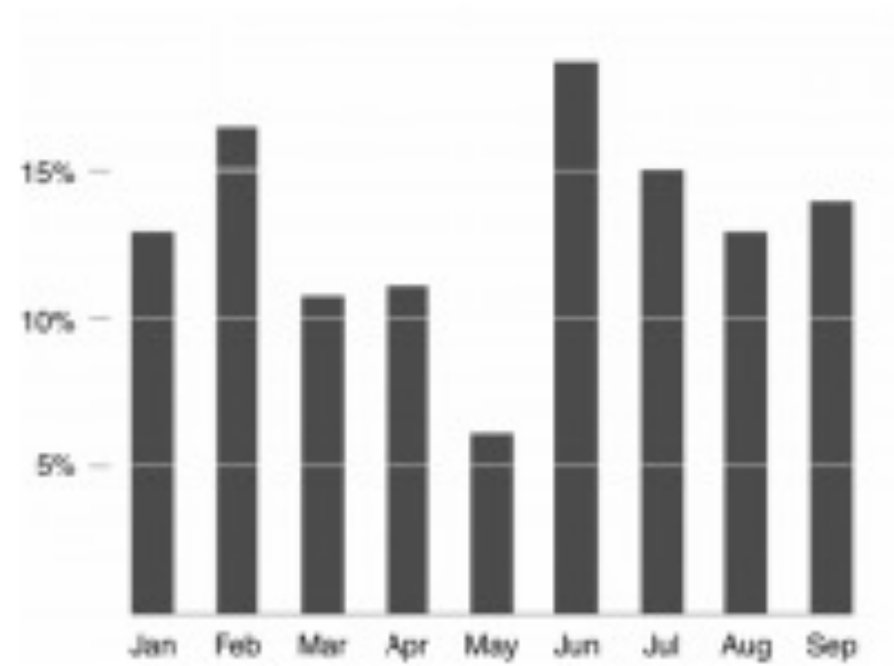
Nightingale Rose / Coxcomb chart



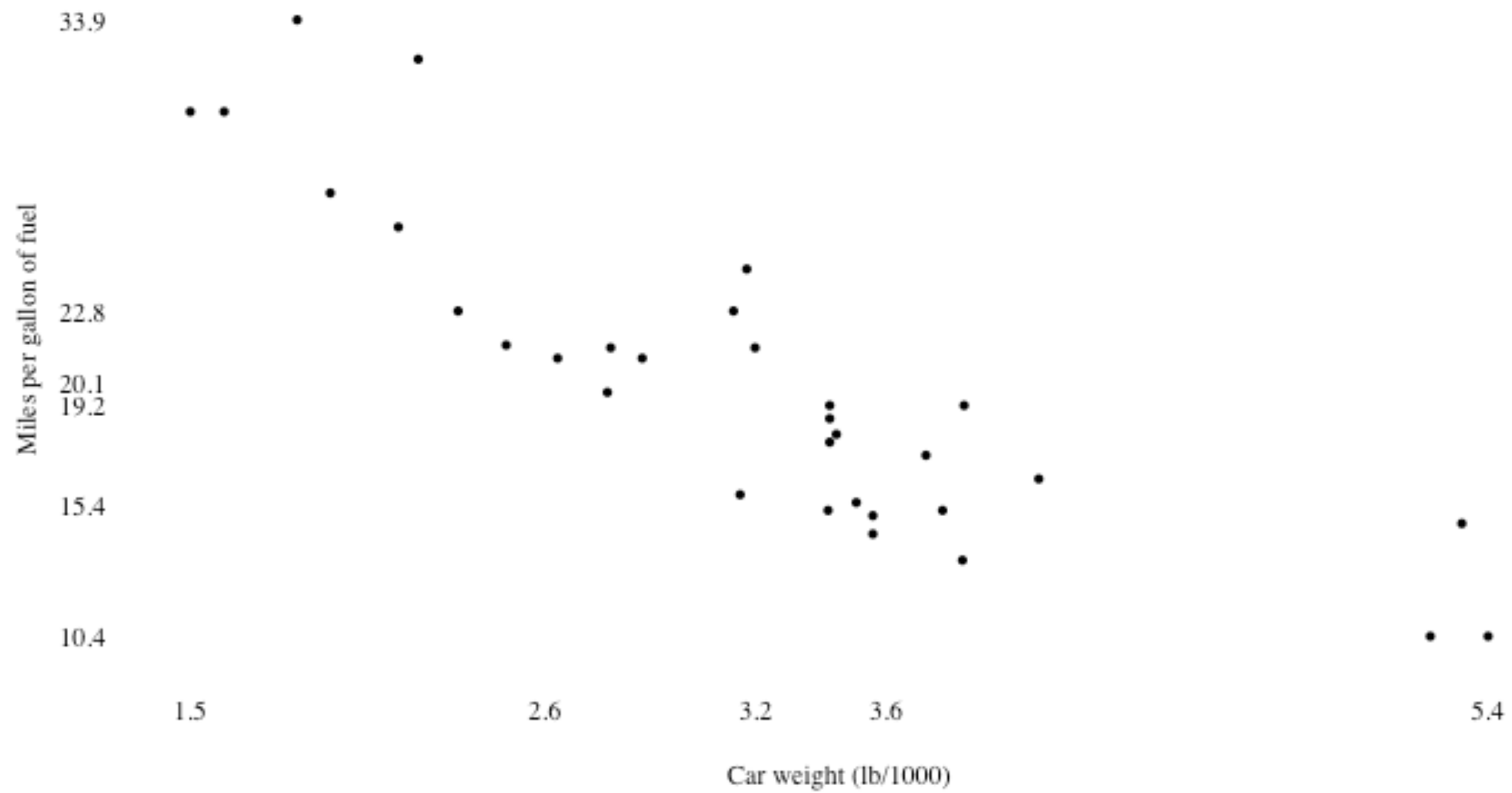


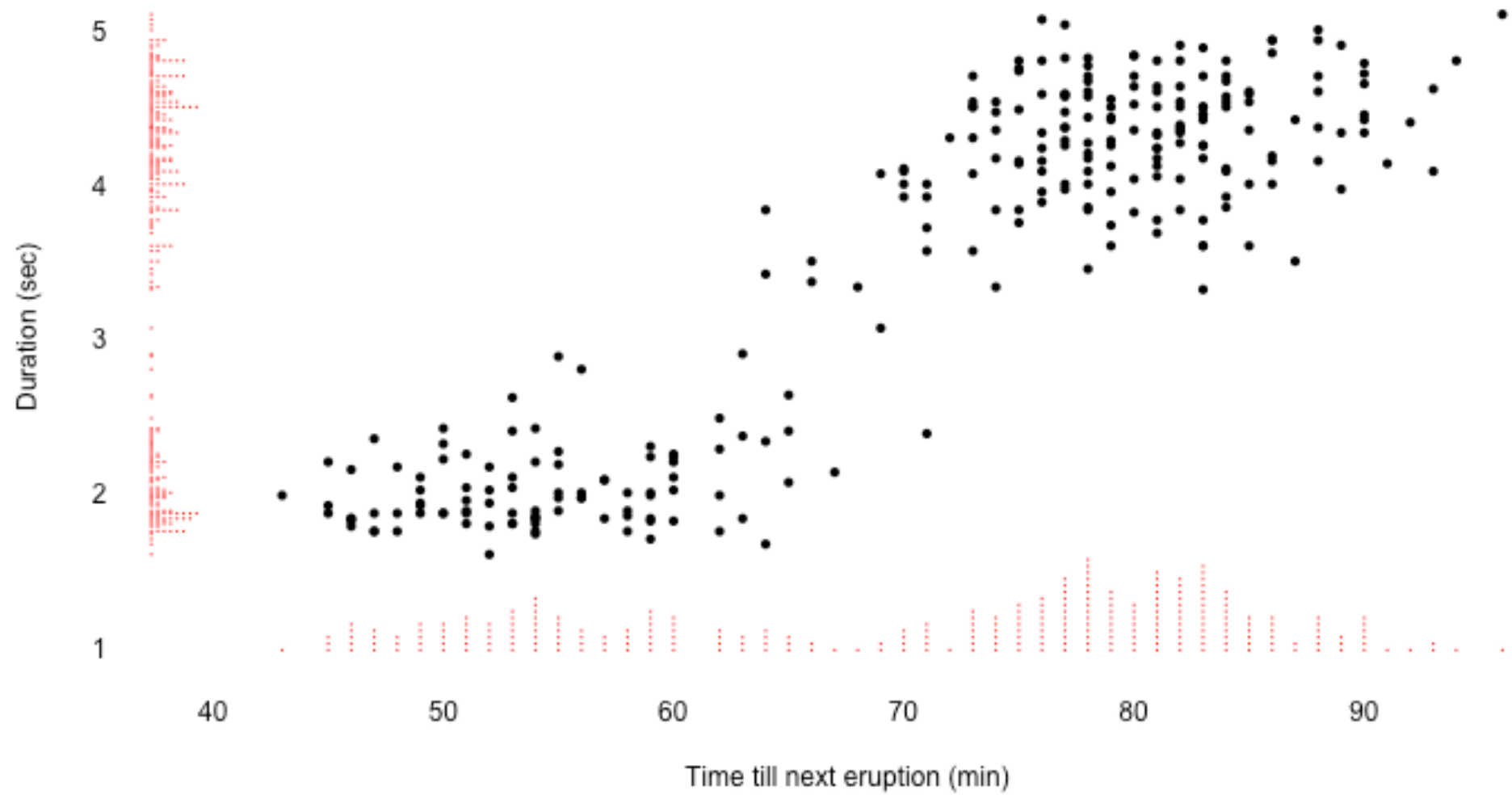


Low Data/Ink

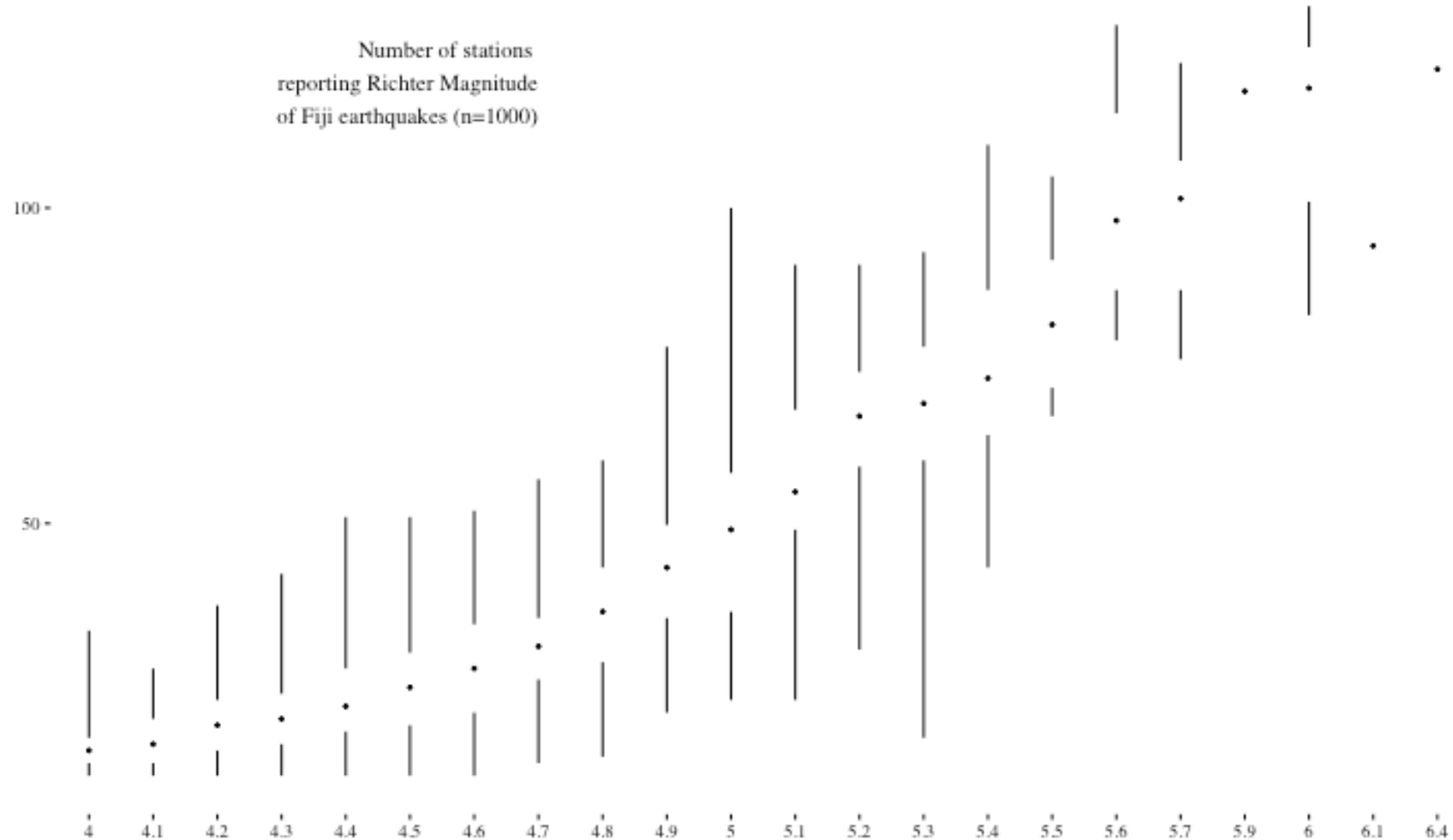


High Data/Ink






```
ggplot(quakes, aes(factor(mag), stations)) +
  theme_tufte() +
  geom_tufteboxplot(outlier.colour = "transparent") +
  theme(axis.title = element_blank())
```



Tufte wisdom

- Tufte's principles are more oriented to communication and can be taken too far
- Better data/ink → display more information without overload;
- Thinking about perception can help you choose better geoms, aesthetics.

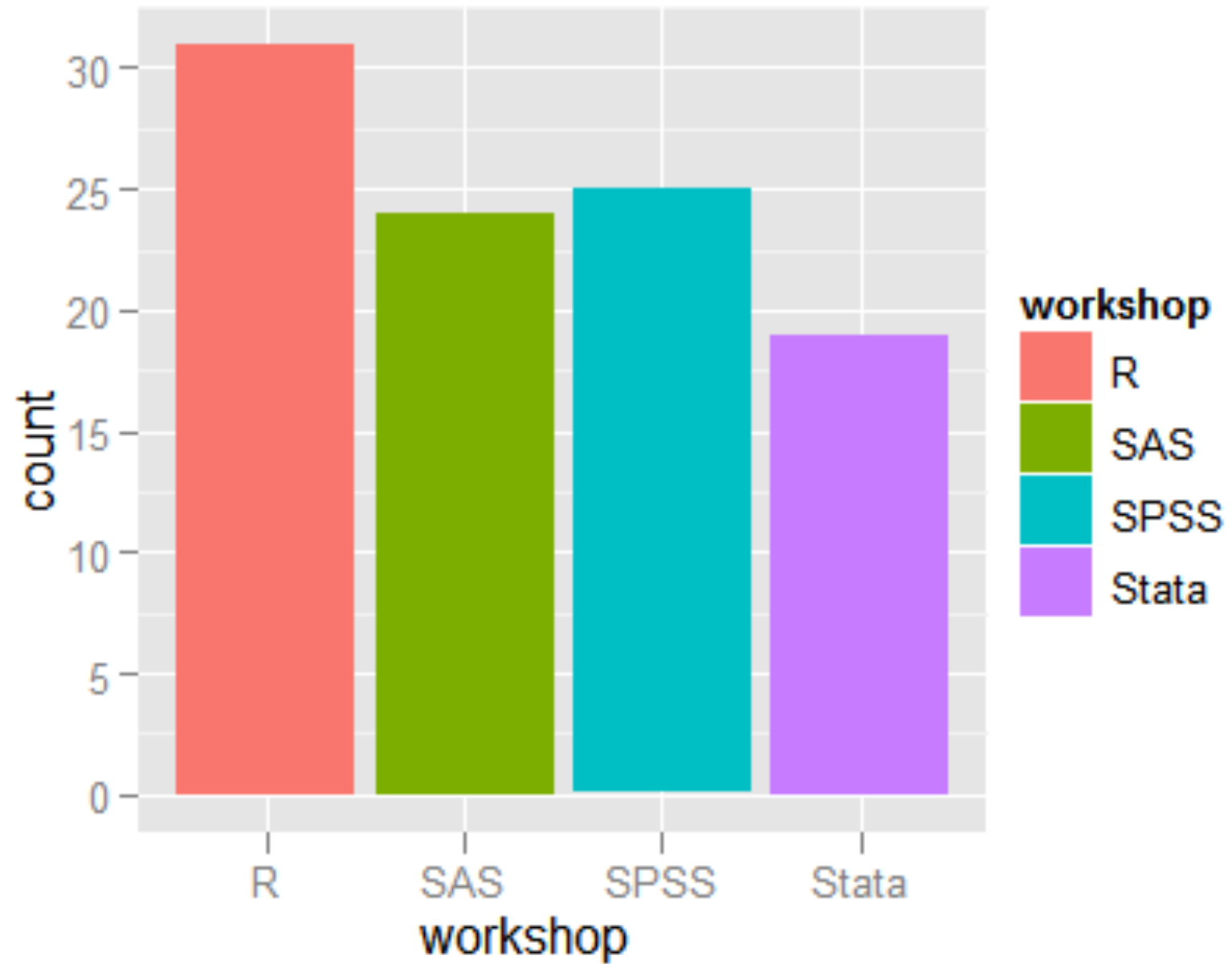


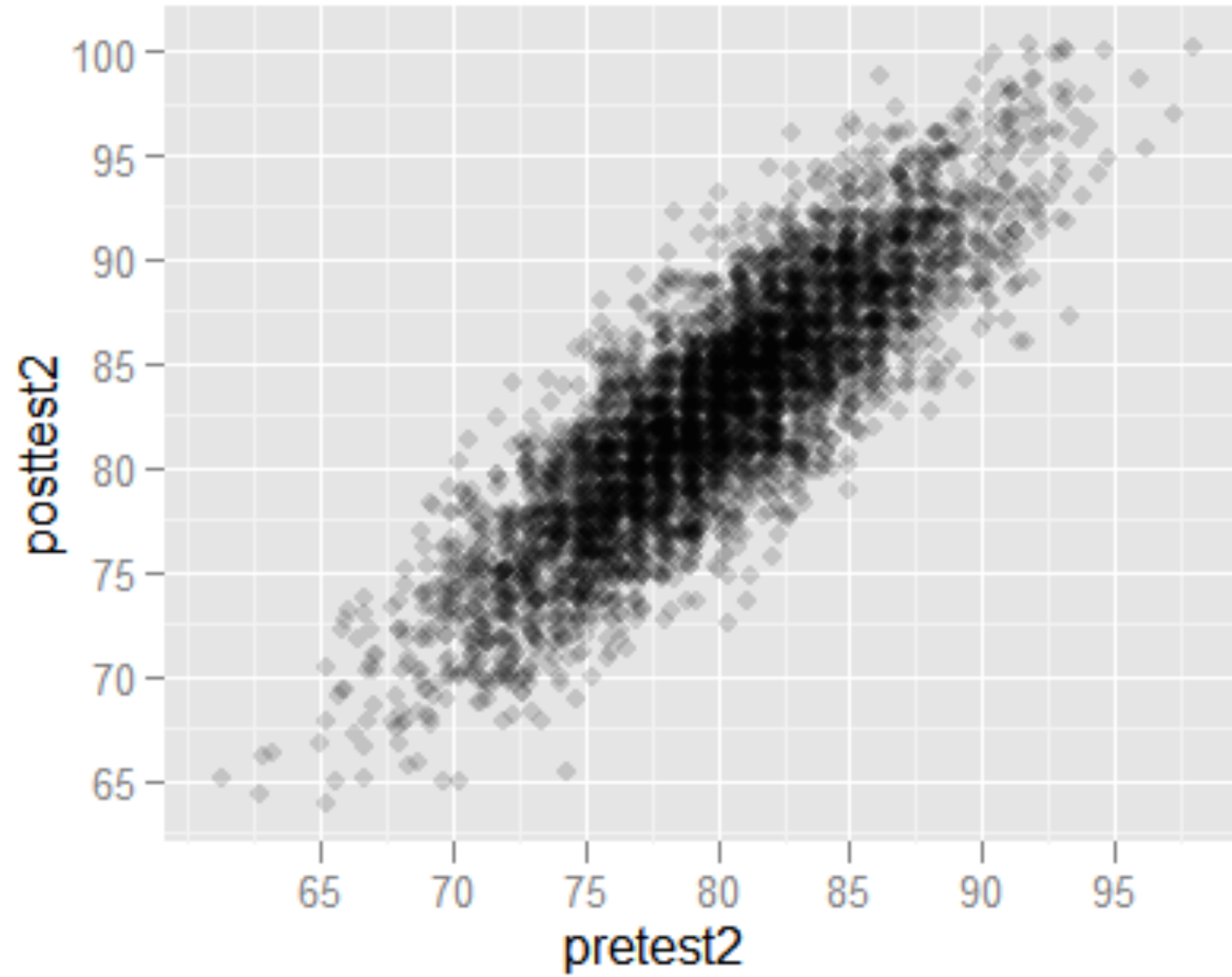
Some practice

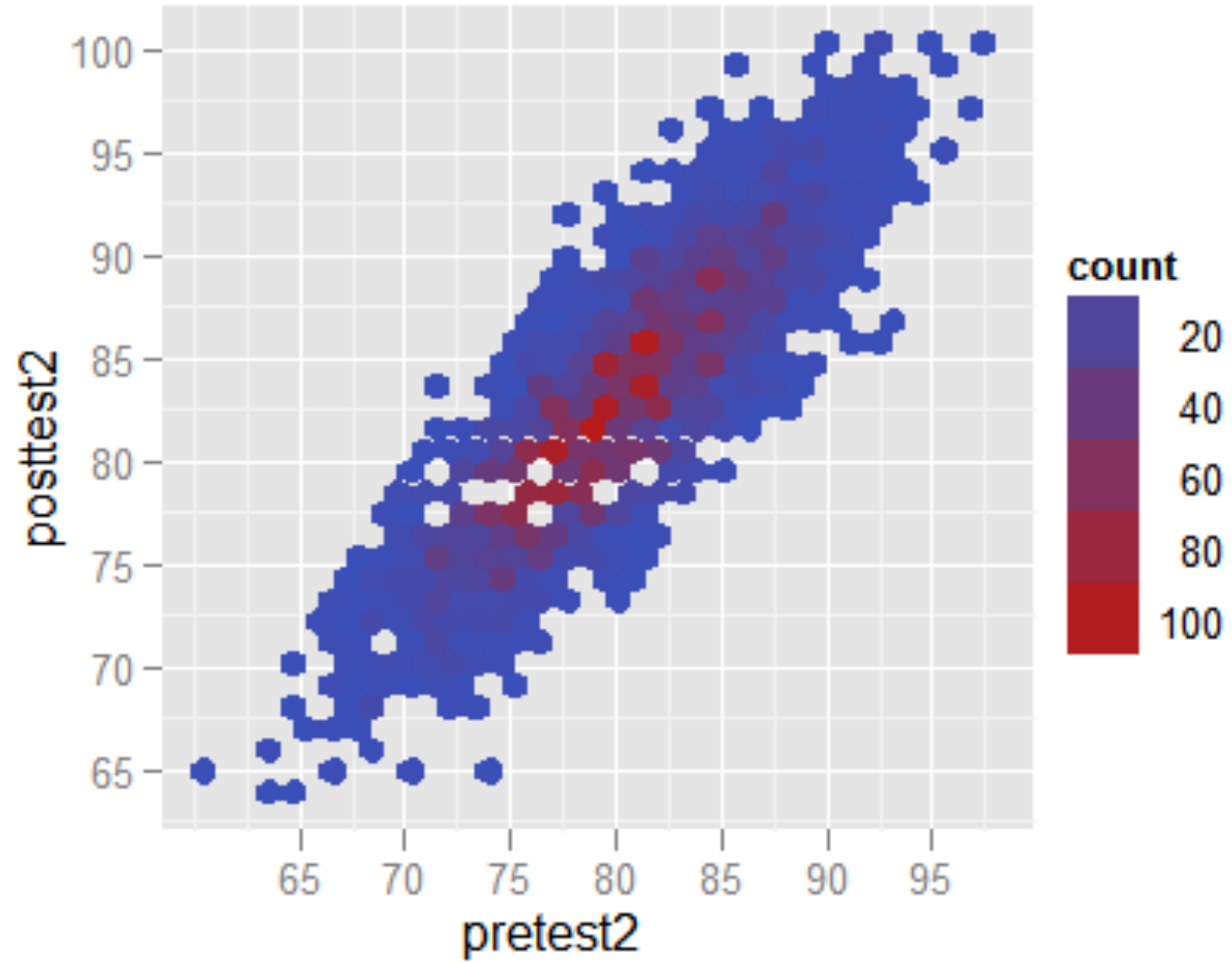
Answer these questions:

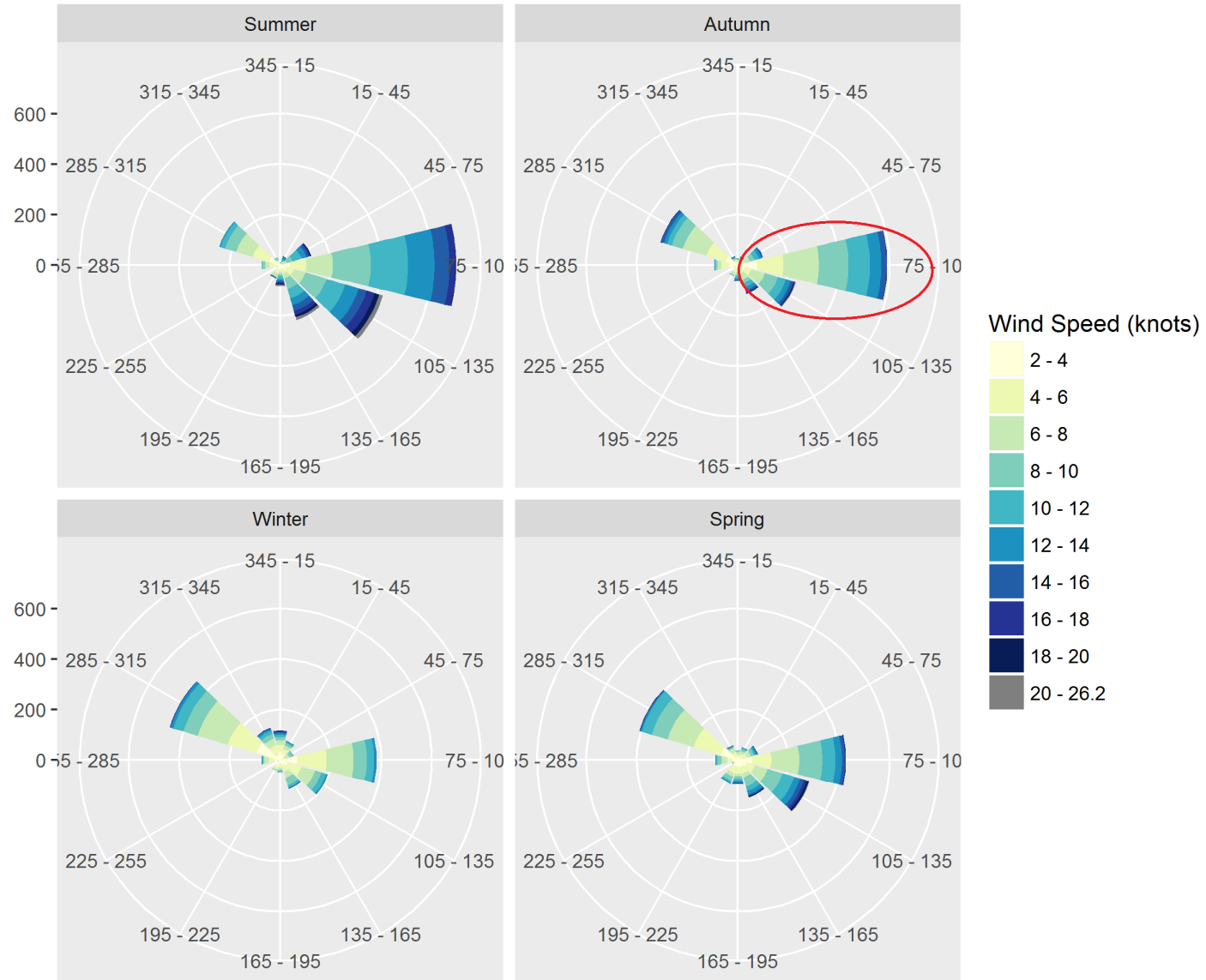
- Are we plotting the right thing?
- What are: aesthetics, geom, scale, facets, transformation, coordinate system
- How is data/ink?
- Is perception considered optimally?
- Can you think of questions you can't answer from this plot which are in the data?











Conclusion

Conclusion

- Data visualization is data analysis + psychology;
- Sticking to **basic principles** helps:
 - **Map data** to aesthetics, geoms, scales, facets;
 - Perception research guides choices;
 - **Which comparisons** do I want?
 - Maximize **data-ink** (within reason).
- Don't believe everything you hear ("do's and don'ts")

